# Spectral Shrunk Clustering: Reproduction, Numerical Audits, and Subset Uncertainty

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## Abstract

Spectral Shrunk Clustering (SSC) refines a spectral representation through a convex fidelity–fusion objective. We separate paper reconstruction, numerical optimization and clustering quality. Joint ACC/NMI matching and declared source/subset sensitivities preserve distinct statistical and identity qualifications. Certified shrinkage has small, mixed net effects at gamma 1; normalization and graph interventions expose different limitations. An oracle completion gate passes on ORL, while label-free four-benchmark improvement remains unestablished. Source and implementation audits retain failures and adapted methods explicitly. Numerical agreement does not identify the authors' historical datasets.

## 1 What SSC optimizes

Spectral clustering (SC) represents samples using graph-Laplacian eigenvectors and then applies K-means. Chang et al. [2015] argue that directly clustering the initial spectral coordinates can be unreliable, and propose learning *shrunk patterns* that preserve their anchors while contracting neighboring representations. Their contribution is a convex representation-refinement formulation, relaxation of an orthogonality constraint in this stage, and an iterative algorithm with clustering experiments. SSC here means Spectral Shrunk Clustering, rather than Sparse Subspace Clustering.

The input-space affinity $A$ determines $L_A = I - D_A^{-1/2} A D_A^{-1/2}$. The $c$ smallest eigenvectors form $F \in \mathbb{R}^{n\times c}$. A second affinity $W$ is constructed from $F$ and controls refinement (Figure 1). With each undirected edge stored once in incidence matrix $B$, Equation 8 is

$$P(G) = \sum_i \|g_i - f_i\|_2 + 2\gamma \sum_e w_e \|(BG)_e\|_2. \tag{1}$$

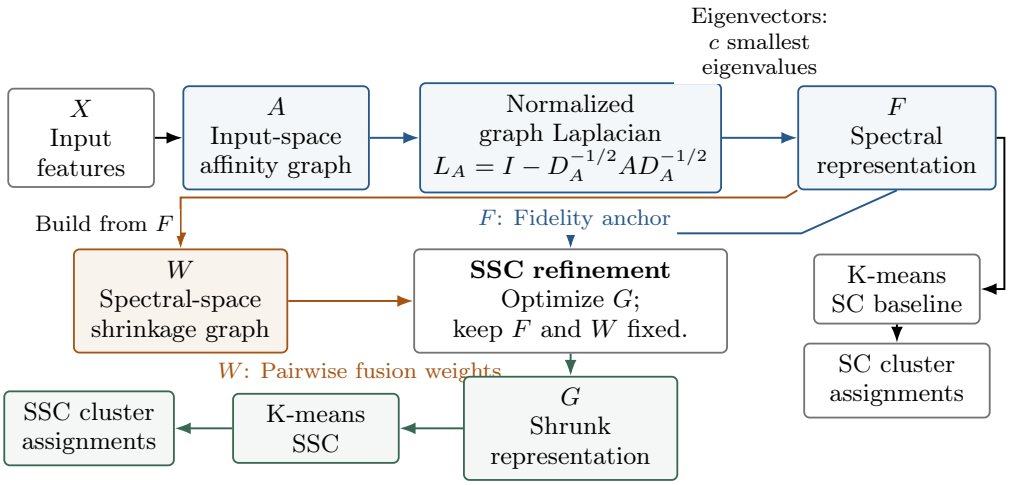
The factor 2 implements the paper's ordered-pair sum. Its stationary IRLS system uses $S + 2\gamma\widetilde{L}$ (E8), whereas printed Equation 13 uses $S + \gamma\widetilde{L}$ (E13). Thus E13 at nominal gamma targets E8 at effective gamma/2. We retain both tracks, rather than calling unequal targets a solver discrepancy.

For fixed $F, W$, nonnegative weights make (1) convex; unsquared fidelity does not guarantee uniqueness. Graph construction, eigenspace selection and final K-means are separate operations. A globally optimal $G$ therefore need not improve ACC or NMI.

## 2 Published results and the reproduction protocol

We verified all 288 reported means and SD values in the paper's Tables 1–2: nine methods, eight benchmarks and two metrics. The paper states $k = 5$, 50 random initializations, a five-value parameter grid and native-objective selection. It does not fully resolve the SD hierarchy, source indices, Gaussian

Preprint.




**SSC refinement objective (Equation 8)**

$$\min_G \Big[\underbrace{\sum_i \|g_i - f_i\|_2}_{\text{Fidelity to } F} + \underbrace{2\gamma \sum_{i<j} W_{ij}\|g_i - g_j\|_2}_{\text{Fusion controlled by } W}\Big]$$

Each undirected edge once; both norms unsquared.

Figure 1: The two graph roles are distinct. $A$ determines the initial spectral geometry $F$; $W$, built in that geometry, controls fusion while $F$ remains fixed. Matched SC clusters the same $F$, and SSC clusters $G$.

width or graph symmetrization. Our choices are union kNN, median positive directed-neighbor squared-distance width, deterministic eigensolver initialization, and explicit one-start Lloyd rounding. ACC uses optimal label assignment and NMI uses geometric entropy normalization.

The paper's SSC ACC gains do not imply superiority on every metric: YaleB SC/SSC have ACC 34.8/49.8 but NMI 55.6/46.8. We therefore minimize joint discrepancy in percentage points,

$$D_{\max} = \max(|\overline{\text{ACC}} - \text{ACC}_p|, |\overline{\text{NMI}} - \text{NMI}_p|), \qquad D_{\text{mean}} = \tfrac{1}{2}(d_{\text{ACC}} + d_{\text{NMI}}).$$

We rank by $D_{\max}$, then $D_{\text{mean}}$, with a $10^{-12}$ tolerance and frozen compatibility tie breaks. Published-grid gamma values are not penalized for differing from 1. Both metrics and SD come from one configuration and its complete 50 seeds. The single native-best-of-50 assignment has no across-run SD; existing independent batches are a separate repeatability endpoint. Raw LDA inertia across learned subspaces is not treated as a common selection objective.

| Data | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ | Fit |
|---|---|---|---|---|---|---|
| AR | $51.30 \pm 1.50$ | $51.21 \pm 2.00$ | $73.20 \pm 1.40$ | $72.94 \pm 1.05$ | 0.26 | T |
| JAFFE | $81.20 \pm 1.60$ | $74.75 \pm 8.37$ | $84.30 \pm 1.60$ | $85.60 \pm 4.32$ | 6.45 | T |
| ORL | $76.00 \pm 1.60$ | $74.63 \pm 0.64$ | $88.60 \pm 1.50$ | $89.71 \pm 0.23$ | 1.37 | I |
| UMIST | $71.10 \pm 1.80$ | $69.97 \pm 2.27$ | $84.10 \pm 1.50$ | $85.17 \pm 1.00$ | 1.13 | T |
| BinAlpha | $49.40 \pm 1.30$ | $49.54 \pm 2.71$ | $64.10 \pm 1.40$ | $63.46 \pm 1.04$ | 0.64 | I |
| MSRA50† | $63.20 \pm 1.10$ | $49.25 \pm 4.38$ | $72.20 \pm 1.40$ | $61.50 \pm 3.89$ | 13.95 | T |
| YaleB | $49.80 \pm 1.60$ | $44.24 \pm 0.82$ | $46.80 \pm 1.30$ | $52.61 \pm 0.36$ | 5.81 | T |
| USPS | $75.50 \pm 1.90$ | $66.26 \pm 1.70$ | $79.80 \pm 1.60$ | $80.86 \pm 1.20$ | 9.24 | I |

Table 1: Closest evaluated SSC reconstructions under joint $D_{\max}/D_{\text{mean}}$ matching. This is retrospective numerical selection using labels and paper targets; historical subset identity remains unverified. Means $\pm$ sample SD use one configuration and all 50 seeds. Compatible-pool winners are shown where available; † marks a separately selected adaptation. I: iterate stopping without a gap certificate; T: iteration cap; U: other uncertified state. Exact candidate IDs, counts, signed differences, settings and both pools appear in the appendix. Historical selections are retained separately.

### 2.1 Dataset-subset uncertainty and retrospective numerical matching

Original sample indices were unavailable. Final source/subset matches and locked-selection rounding are reported separately from the historical and method-comparison endpoints. **Such matching is retrospective, not label-free selection or an unbiased superiority test.** Exact manifests support replay, not historical identity. Changed membership rebuilds $A, F, W, G$; source, resize, normalization and eligibility pools remain distinct (Appendix C).

The evaluated design obtained complete SSC configurations on 2482 unique inputs, yielding 74436 input/configuration pairs. Of these, 2480 inputs completed all 30 SSC settings and 2480 completed the 30-SSC plus three matched-SC design. Aliases are provenance entries, and diagnostic parents are excluded from equivalent paper reconstruction. Additional rounding on locked selections measures conditional stability; closer numerical matching does not identify historical data.

## 3 What discrepancies can be explained

**Numerics.** For $c_e = 2\gamma w_e$, feasible $q$ with $\|q_e\| \leq c_e$ and $\|(B^T q)_i\| \leq 1$ gives lower bound $D(q) = \langle B^T q, F\rangle$. We retain the initial feasible bound in preconditioned PDHG [Chambolle and Pock, 2011]; certification requires relative gap at most $10^{-5}$ and capacity violations at most $10^{-8}$. On one frozen current USPS $F/W$, the gamma-1 continuation certifies its saved primal at gap $9.9996 \times 10^{-6}$, yet ACC/NMI change by only $-0.033/-0.086$ points against same-$F$ SC. Gamma 1000 remains uncertified at gap 0.342; its collapsed saved point does not prove optimal collapse. Failed IRLS/dual-flow attempts and missing historical arrays remain separate limitations.

**Implementation and sources.** The pinned ClustPy implementation of LDA-Kmeans [Ding and Li, 2007] failed every LDA update in all 50 ORL and JAFFE fits; these are implementation-specific PCA–K-means degeneracies. The projection actually clustered and the source-returned projection are recorded separately. Existing-project LDA limits 100/300 give identical per-seed ACC/NMI on all five tested sources. The separately named ridge LDA-Kmeans→SC adaptation improves both means over raw/PCA$_{p_0}$ SC only on YaleB; it does not recover historical LDA-SC. Paired SLPS [Hou et al., 2013] $k = 5/7$ effects change with source and alpha. Native 16-by-16 EBSA MSRA retains all 1,799 rows and duplicates. Exact pins, adaptation formulas and controls are in Appendix C.

The baseline comparison contains 98 complete 50-seed SLPS configurations, 30 LDA configurations/control endpoints and 62 other configurations. The appendix retains numerical failures, iteration-limit outcomes and update diagnostics. Native 16-by-16 MSRA and ridge LDA-to-SC are explicitly named adaptations; exact historical LDA-SC remains unresolved.

## 4 Graph diagnostics and conditional extension

G1 compares SC/SSC on the same fixed representation. Certified gamma-1 shrinkage changes mean ACC/NMI by $+0.155/+0.014$ points on ORL and $-0.270/-0.276$ on YaleB, despite active movement. $A$ and $W$ have different roles: cross-class edges carry 4.6% of ORL $W$ mass but 70.2% of YaleB $W$ mass. Class-induced components, including isolates, distinguish within-class fragmentation from whole-graph connectivity. YaleB's $W$ edge-length AUROC is 0.380, weakening the general premise that erroneous edges are longer; these associations alone do not establish their causal effect.

G2 separates anchor normalization from the fusion graph at matched mass. Row-normalized $U$ alone raises SC ACC by 1.850/4.016 points on ORL/YaleB; angular-graph shrinkage at gamma 1 adds only 0.170/0.314 above $U$-SC. A stronger representation baseline therefore matters more than a comparison only with $F_0$-SC.

In the fixed G3 oracle intervention (15% additions, gamma 1000), ORL passes the diagnostic criterion: completion gains 10.415 ACC points over deletion and 24.863 over fixed oracle deletion plus mean blind additions, with nondecreasing NMI and qualified conditional intervals. YaleB's large raw gains remain numerically unqualified. The EnSC graph adaptation [You et al., 2016] is evaluated separately: oracle success does not establish an unsupervised-method gain. Native EnSC clustering and matched-movement controls distinguish affinity quality from the additional effect of SSC fusion.

The fixed oracle graph-intervention criterion is passed. The capability and locked-operation macro targets are unavailable because all four required comparisons must be qualified. Oracle graph evidence, unsupervised affinity construction and net shrinkage gains are distinct conclusions.

The primary EnSC–SSC capability means are 78.98/96.41/27.66/65.60 ACC on ORL/JAFFE/YaleB/USPS. The first three select gamma 1, but USPS selects gamma 0.001 with $G = F$: this is damage avoidance, not useful shrinkage. USPS at the fixed gamma 1 falls to 52.27 ACC and remains uncertified. Native EnSC has higher ACC/NMI than the primary adaptation on the first three datasets; the USPS primary exceeds native EnSC without moving from $F_0$.

The target is at least 2 points equal-weight ACC gain over the stronger fair SC/SSC baseline, with no mean NMI loss against either. Fair SC includes both $F_0$ and $U$. All four datasets have complete 50-start records, but required uncertified controls leave both capability and locked-operation macro targets unavailable. The parameter-selection-gap flag is therefore undefined. Additional seeds assess conditional rounding stability, not unseen-data generalization.

## 5 Conclusion

SSC gives a clear convex refinement target, while clustering quality depends on representation, graph, solver and rounding choices. Our contribution is reproducible source/subset sensitivity and explicit numerical and implementation audits, including negative or unresolved evidence. Exact historical LDA-SC and several dataset identities remain unresolved. Code, statistical records and reproducible commands are available in the public repository.

## References


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# A Historical measurements and endpoint separation

The original scalar measurements and ACC-only selections remain distinct historical endpoints. Joint selections use complete seed groups and preserve their original measurements. Missing historical representations limit optimization re-audits on their exact arrays. The historical 72-cell identity/status matrix remains separate from newly measured candidates and adaptations. The original ACC-only closest-display and fixed-reference tables remain available in the committed historical release. The current joint 72-cell table below supersedes their display selection without rewriting any trial.

# B Results, statistical endpoints and coverage

## B.1 Jointly reselected historical mean endpoint

Table 2 reports the complete 72-cell inventory. Values are percent. Every measured pair uses the same configuration and 50 batch-zero seeds; $\pm$ denotes sample SD. Missing cells retain their identity or method restriction. Historical scalar evidence is separate from new recoverable artifacts. The former ACC-only and fixed-reference tables remain in the committed historical release.

Table 2: Joint closest historical mean/sample-SD endpoint; unavailable cells are not filled with adapted measurements.

| Data | Method | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| AR | K-means | $36.30 \pm 1.40$ | – | $68.70 \pm 3.00$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | SC | $41.60 \pm 2.10$ | – | $71.30 \pm 2.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | PCA–K-means | $39.80 \pm 1.80$ | – | $69.40 \pm 2.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | PCA–SC | $43.20 \pm 1.70$ | – | $70.30 \pm 2.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | LDA–K-means | $40.40 \pm 1.50$ | – | $69.90 \pm 1.90$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | LDA–SC | $44.50 \pm 1.30$ | – | $70.80 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | LLC | $48.70 \pm 1.60$ | – | $71.20 \pm 2.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | SLPS | $49.20 \pm 1.40$ | – | $72.40 \pm 1.70$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| AR | SSC | $51.30 \pm 1.50$ | – | $73.20 \pm 1.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| JAFFE | K-means | $75.60 \pm 1.80$ | $71.97 \pm 7.55$ | $79.40 \pm 0.80$ | $81.97 \pm 3.70$ | 3.63 |
| JAFFE; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 3.10$. | | | | | | |
| JAFFE | SC | $76.10 \pm 1.60$ | $68.53 \pm 11.15$ | $80.20 \pm 0.90$ | $83.02 \pm 5.72$ | 7.57 |
| JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 5.20$. | | | | | | |
| JAFFE | PCA–K-means | $75.80 \pm 1.50$ | $73.20 \pm 5.58$ | $79.80 \pm 0.80$ | $82.95 \pm 3.67$ | 3.15 |
| JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.88$. | | | | | | |
| JAFFE | PCA–SC | $76.90 \pm 1.80$ | $72.30 \pm 0.00$ | $81.50 \pm 1.30$ | $82.86 \pm 0.00$ | 4.60 |
| JAFFE; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.98$. | | | | | | |
| JAFFE | LDA–K-means | $76.50 \pm 1.70$ | $64.44 \pm 9.19$ | $82.10 \pm 1.40$ | $78.60 \pm 5.08$ | 12.06 |
| JAFFE; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 7.78$. | | | | | | |
| JAFFE | LDA–SC | $77.40 \pm 1.90$ | – | $81.90 \pm 0.90$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| JAFFE | LLC | $78.60 \pm 1.50$ | $91.55 \pm 0.00$ | $82.50 \pm 1.70$ | $92.75 \pm 0.00$ | 12.95 |
| JAFFE; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 11.60$. | | | | | | |
| JAFFE | SLPS | $79.50 \pm 2.10$ | $80.86 \pm 5.77$ | $83.10 \pm 2.10$ | $86.66 \pm 3.34$ | 3.56 |
| JAFFE; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.46$. | | | | | | |
| JAFFE | SSC | $81.20 \pm 1.60$ | $74.75 \pm 8.37$ | $84.30 \pm 1.60$ | $85.60 \pm 4.32$ | 6.45 |
| JAFFE; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $D_{\text{mean}} = 3.87$. | | | | | | |
| ORL | K-means | $60.50 \pm 1.80$ | $61.46 \pm 2.85$ | $80.30 \pm 1.80$ | $82.76 \pm 1.17$ | 2.46 |
| ORL_att92; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 1.71$. | | | | | | |
| ORL | SC | $72.70 \pm 2.30$ | $72.73 \pm 3.15$ | $85.80 \pm 1.90$ | $87.49 \pm 1.22$ | 1.69 |
| ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 0.86$. | | | | | | |
| ORL | PCA–K-means | $64.50 \pm 2.30$ | $63.08 \pm 2.26$ | $80.60 \pm 1.60$ | $82.34 \pm 0.90$ | 1.74 |
| ORL_att92; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 1.58$. | | | | | | |

| Data | Method | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| ORL | PCA–SC | $67.80 \pm 2.10$ | $67.57 \pm 2.92$ | $86.30 \pm 1.40$ | $85.81 \pm 1.27$ | 0.49 |
| ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.36$. | | | | | | |
| ORL | LDA–K-means | $65.60 \pm 2.60$ | $56.31 \pm 4.14$ | $81.10 \pm 2.10$ | $81.00 \pm 1.65$ | 9.29 |
| ORL_att92; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\mathrm{mean}} = 4.70$. | | | | | | |
| ORL | LDA–SC | $68.30 \pm 2.40$ | – | $86.80 \pm 1.70$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| ORL | LLC | $71.50 \pm 2.20$ | $81.07 \pm 0.61$ | $84.90 \pm 1.50$ | $89.92 \pm 0.27$ | 9.57 |
| ORL_att92; Y0, YU_SHI; NOT_APPLICABLE; $D_{\mathrm{mean}} = 7.29$. | | | | | | |
| ORL | SLPS | $74.20 \pm 1.80$ | $73.66 \pm 2.13$ | $87.20 \pm 1.80$ | $87.71 \pm 0.91$ | 0.54 |
| ORL_att92; Y0, one_shot, G; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.53$. | | | | | | |
| ORL | SSC | $76.00 \pm 1.60$ | $74.63 \pm 0.64$ | $88.60 \pm 1.50$ | $89.71 \pm 0.23$ | 1.37 |
| ORL_att92; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $D_{\mathrm{mean}} = 1.24$. | | | | | | |
| UMIST | K-means | $41.30 \pm 1.60$ | – | $64.40 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | SC | $52.20 \pm 1.40$ | – | $72.10 \pm 1.70$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | PCA–K-means | $48.80 \pm 1.70$ | – | $68.20 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | PCA–SC | $54.10 \pm 1.90$ | – | $72.90 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | LDA–K-means | $49.70 \pm 1.80$ | – | $68.80 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | LDA–SC | $54.70 \pm 1.50$ | – | $74.10 \pm 2.00$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | LLC | $63.30 \pm 1.80$ | – | $77.30 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | SLPS | $70.40 \pm 1.60$ | – | $82.20 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| UMIST | SSC | $71.10 \pm 1.80$ | – | $84.10 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | K-means | $41.70 \pm 1.10$ | – | $58.60 \pm 1.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | SC | $43.60 \pm 1.50$ | – | $59.70 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | PCA–K-means | $42.40 \pm 1.50$ | – | $59.10 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | PCA–SC | $44.30 \pm 1.80$ | – | $60.60 \pm 1.90$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | LDA–K-means | $42.90 \pm 1.70$ | – | $59.80 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | LDA–SC | $45.10 \pm 1.40$ | – | $61.30 \pm 1.70$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | LLC | $40.70 \pm 1.80$ | – | $61.40 \pm 1.90$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | SLPS | $48.50 \pm 1.90$ | – | $63.60 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| BinAlpha | SSC | $49.40 \pm 1.30$ | – | $64.10 \pm 1.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | K-means | $46.20 \pm 1.70$ | – | $56.70 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | SC | $52.30 \pm 1.80$ | – | $70.00 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | PCA–K-means | $56.00 \pm 1.90$ | – | $60.30 \pm 1.50$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | PCA–SC | $54.90 \pm 1.60$ | – | $72.40 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | LDA–K-means | $56.40 \pm 1.80$ | – | $61.10 \pm 1.90$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | LDA–SC | $55.10 \pm 1.70$ | – | $73.20 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | LLC | $48.10 \pm 1.40$ | – | $66.20 \pm 1.60$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | SLPS | $60.30 \pm 1.30$ | – | $69.60 \pm 1.80$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| MSRA50 | SSC | $63.20 \pm 1.10$ | – | $72.20 \pm 1.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |

| Data | Method | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| YaleB | K-means | $14.40 \pm 1.50$ | $9.74 \pm 0.46$ | $17.30 \pm 1.50$ | $13.25 \pm 0.59$ | 4.66 |
| YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.35$. | | | | | | |
| YaleB | SC | $34.80 \pm 1.40$ | $37.39 \pm 2.22$ | $55.60 \pm 1.60$ | $52.96 \pm 1.00$ | 2.64 |
| YaleB_dengcai32; Y3, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.61$. | | | | | | |
| YaleB | PCA–K-means | $24.90 \pm 1.80$ | $9.85 \pm 0.55$ | $26.70 \pm 1.80$ | $13.32 \pm 0.66$ | 15.05 |
| YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 14.22$. | | | | | | |
| YaleB | PCA–SC | $36.80 \pm 2.00$ | $30.92 \pm 0.54$ | $38.60 \pm 1.50$ | $38.86 \pm 0.95$ | 5.88 |
| YaleB_dengcai32; Y1, G; NOT_APPLICABLE; $D_{\text{mean}} = 3.07$. | | | | | | |
| YaleB | LDA–K-means | $26.10 \pm 1.80$ | $9.80 \pm 0.85$ | $29.40 \pm 1.60$ | $13.79 \pm 1.15$ | 16.30 |
| YaleB_dengcai32; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 15.96$. | | | | | | |
| YaleB | LDA–SC | $38.20 \pm 1.60$ | – | $39.90 \pm 1.40$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| YaleB | LLC | $38.20 \pm 1.50$ | $32.35 \pm 0.05$ | $34.10 \pm 1.30$ | $39.68 \pm 0.11$ | 5.85 |
| YaleB_dengcai32; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 5.72$. | | | | | | |
| YaleB | SLPS | $47.30 \pm 1.70$ | $13.52 \pm 0.78$ | $41.40 \pm 1.60$ | $19.44 \pm 0.69$ | 33.78 |
| YaleB_dengcai32; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 27.87$. | | | | | | |
| YaleB | SSC | $49.80 \pm 1.60$ | $42.64 \pm 0.61$ | $46.80 \pm 1.30$ | $52.19 \pm 0.31$ | 7.16 |
| YaleB_dengcai32; Y3, E8, $\gamma = 1e+06$, G; MAX_ITER_REACHED; $D_{\text{mean}} = 6.27$. | | | | | | |
| USPS | K-means | $65.40 \pm 1.70$ | $66.06 \pm 2.17$ | $67.30 \pm 1.80$ | $61.04 \pm 0.87$ | 6.26 |
| USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 3.46$. | | | | | | |
| USPS | SC | $64.30 \pm 1.40$ | $61.74 \pm 8.09$ | $69.50 \pm 1.60$ | $74.94 \pm 4.49$ | 5.44 |
| USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 4.00$. | | | | | | |
| USPS | PCA–K-means | $69.40 \pm 1.80$ | $65.25 \pm 2.85$ | $73.10 \pm 1.90$ | $61.10 \pm 0.95$ | 12.00 |
| USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 8.08$. | | | | | | |
| USPS | PCA–SC | $69.10 \pm 1.50$ | $65.53 \pm 3.26$ | $74.20 \pm 1.80$ | $78.92 \pm 2.12$ | 4.72 |
| USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.15$. | | | | | | |
| USPS | LDA–K-means | $70.10 \pm 1.30$ | $58.61 \pm 6.33$ | $75.10 \pm 1.60$ | $60.29 \pm 4.21$ | 14.81 |
| USPS; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 13.15$. | | | | | | |
| USPS | LDA–SC | $70.40 \pm 1.50$ | – | $75.40 \pm 1.70$ | – | – |
| No eligible complete measured configuration; historical restrictions remain | | | | | | |
| USPS | LLC | $63.90 \pm 1.70$ | $65.27 \pm 0.01$ | $67.50 \pm 1.50$ | $72.76 \pm 0.01$ | 5.26 |
| USPS; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 3.32$. | | | | | | |
| USPS | SLPS | $71.40 \pm 1.60$ | $70.81 \pm 8.33$ | $76.50 \pm 1.80$ | $76.30 \pm 2.47$ | 0.59 |
| USPS; Y0, one_shot, R; NOT_APPLICABLE; $D_{\text{mean}} = 0.39$. | | | | | | |
| USPS | SSC | $75.50 \pm 1.90$ | $73.87 \pm 7.53$ | $79.80 \pm 1.60$ | $82.84 \pm 3.96$ | 3.04 |
| USPS; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $D_{\text{mean}} = 2.34$. | | | | | | |

### B.2 Native-objective-selected endpoint

Within one fixed G, native K-means SSE selects one of 50 assignments. This is a single assignment with no across-run SD. It does not compete with the mean endpoint. Full method-specific endpoint records are in `results/submission_refresh/endpoint_comparison.csv`; LDA cross-subspace native selection is unavailable.

| SSC data | ACC | NMI | $D_{\max}$ | $D_{\text{mean}}$ | Seed | Fit |
|---|---|---|---|---|---|---|
| AR | – | – | – | – | – | – |
| JAFFE | 95.31 | 94.09 | 14.11 | 11.95 | 13 | T |
| ORL | 75.00 | 88.60 | 1.00 | 0.50 | 40 | I |
| UMIST | – | – | – | – | – | – |
| BinAlpha | – | – | – | – | – | – |
| MSRA50 | – | – | – | – | – | – |
| YaleB | 43.33 | 54.15 | 7.35 | 6.91 | 45 | I |
| USPS | 79.62 | 85.38 | 5.58 | 4.85 | 38 | I |

### B.3 Separate historical repeatability batches

Existing multi-batch records are a separate endpoint. Each batch contains 50 disjoint rounding seeds on its fixed input. The following ranges describe variation of batch means and native-selected assignments, without treating a single native assignment as having a within-batch SD. They are not independent datasets, and are not new campaign measurements. All methods and exact configuration IDs are preserved in `historical_repeatability_batches.csv` and its summary/manifest. LDA source-native selection remains diagnostic only.

| SSC setting | Batches | Batch ACC | Batch NMI | Native ACC | Native NMI |
|---|---|---|---|---|---|
| BinAlpha_candidate26x30 / E13 / gamma=1e+00 / R | 10 | 42.21–43.23 | 56.50–57.12 | 44.62–47.82 | 57.53–59.66 |
| BinAlpha_candidate26x30 / E13 / gamma=1e+00 / G | 10 | 47.05–47.52 | 58.55–58.91 | 45.51–49.36 | 58.37–59.70 |
| BinAlpha_candidate26x30 / E8 / gamma=1e+00 / R | 10 | 40.54–41.55 | 55.87–56.61 | 43.59–45.90 | 57.04–59.18 |
| BinAlpha_candidate26x30 / E8 / gamma=1e+00 / G | 10 | 46.35–47.18 | 58.33–59.09 | 45.38–48.33 | 57.06–59.31 |
| JAFFE / E13 / gamma=1e+00 / R | 10 | 61.13–68.59 | 78.60–83.04 | 80.75–95.31 | 89.37–94.16 |
| JAFFE / E13 / gamma=1e+00 / G | 10 | 95.06–95.31 | 94.05–94.14 | 95.31–95.31 | 94.12–94.12 |
| JAFFE / E8 / gamma=1e+00 / R | 10 | 61.08–68.43 | 78.70–83.05 | 80.75–95.31 | 89.37–94.16 |
| JAFFE / E8 / gamma=1e+00 / G | 10 | 94.85–95.31 | 93.99–94.14 | 95.31–95.31 | 94.12–94.12 |
| ORL_att92 / E13 / gamma=1e+00 / R | 10 | 55.83–57.48 | 80.42–81.43 | 63.00–73.25 | 82.97–87.94 |
| ORL_att92 / E13 / gamma=1e+00 / G | 10 | 78.04–78.77 | 89.52–89.86 | 78.25–80.50 | 89.83–90.93 |
| ORL_att92 / E8 / gamma=1e+00 / R | 10 | 56.12–58.05 | 80.70–81.68 | 62.75–70.50 | 83.63–87.91 |
| ORL_att92 / E8 / gamma=1e+00 / G | 10 | 78.25–78.75 | 89.66–89.95 | 78.25–80.50 | 89.97–91.39 |
| UMIST_precropped575 / E13 / gamma=1e+00 / R | 10 | 54.34–57.24 | 75.69–77.00 | 56.70–67.48 | 76.40–82.39 |
| UMIST_precropped575 / E13 / gamma=1e+00 / G | 10 | 59.98–61.01 | 78.21–78.68 | 60.70–60.87 | 77.83–78.02 |
| UMIST_precropped575 / E8 / gamma=1e+00 / R | 10 | 54.49–56.87 | 74.95–76.16 | 54.61–66.61 | 76.62–81.01 |
| UMIST_precropped575 / E8 / gamma=1e+00 / G | 10 | 60.35–61.78 | 78.35–78.90 | 60.35–61.04 | 77.92–78.52 |
| USPS / E13 / gamma=1e+00 / R | 10 | 59.71–62.17 | 74.34–75.71 | 65.47–65.59 | 80.97–81.19 |
| USPS / E13 / gamma=1e+00 / G | 10 | 64.89–66.26 | 79.94–80.86 | 65.48–65.50 | 80.99–81.01 |
| USPS / E8 / gamma=1e+00 / R | 10 | 59.56–61.93 | 74.38–75.53 | 65.63–65.63 | 81.17–81.24 |
| USPS / E8 / gamma=1e+00 / G | 10 | 64.76–65.94 | 79.77–80.89 | 65.62–65.63 | 81.17–81.19 |
| YaleB_cropped2414 / E13 / gamma=1e+00 / R | 10 | 17.96–18.20 | 26.41–26.67 | 19.51–21.29 | 26.78–27.84 |
| YaleB_cropped2414 / E13 / gamma=1e+00 / G | 10 | 21.59–21.83 | 27.49–27.75 | 21.71–21.96 | 27.41–28.23 |
| YaleB_cropped2414 / E8 / gamma=1e+00 / R | 10 | 17.46–17.89 | 26.05–26.34 | 18.77–20.34 | 25.94–28.07 |
| YaleB_cropped2414 / E8 / gamma=1e+00 / G | 10 | 21.40–21.56 | 27.24–27.58 | 21.25–22.08 | 27.07–28.19 |
| YaleB_dengcai32 / E13 / gamma=1e+00 / R | 10 | 19.80–20.29 | 29.21–29.63 | 21.46–23.28 | 28.80–31.77 |
| YaleB_dengcai32 / E13 / gamma=1e+00 / G | 10 | 25.32–25.55 | 31.13–31.47 | 24.98–25.52 | 31.31–31.96 |
| YaleB_dengcai32 / E8 / gamma=1e+00 / R | 10 | 19.50–19.88 | 29.09–29.32 | 20.59–23.16 | 29.52–30.98 |
| YaleB_dengcai32 / E8 / gamma=1e+00 / G | 10 | 25.31–25.46 | 31.24–31.43 | 25.02–25.10 | 31.54–31.65 |

### B.4 Dataset-subset matching: both endpoints and eligibility pools

The following rows use complete evaluated configurations only. Paper values are fixed SSC targets. R/V/G denote random, vanilla K-means++ and sklearn greedy K-means++; each is one Lloyd start. Specification-compatible dimensions/counts do not establish historical pixel identity. Adapted rows retain explicit source/resolution/preparation differences. Diagnostic parents cannot win either pool.

**Mean/sample-SD over 50 fixed-G roundings**

| Data / pool | Selected candidate | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| AR / compatible | AR_common_mask0092 | $51.30 \pm 1.50$ | $51.21 \pm 2.00$ | $73.20 \pm 1.40$ | $72.94 \pm 1.05$ | 0.26 |

$n = 840$; Y0, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -0.09$ pp; $\Delta_{\mathrm{NMI}} = -0.26$ pp; $D_{\mathrm{mean}} = 0.18$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.50$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.35$ pp.

| Data / pool | Selected candidate | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| JAFFE / compatible | JAFFE | $81.20 \pm 1.60$ | $74.75 \pm 8.37$ | $84.30 \pm 1.60$ | $85.60 \pm 4.32$ | 6.45 |
| $n = 213$; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -6.45$ pp; $\Delta_{\mathrm{NMI}} = 1.30$ pp; $D_{\mathrm{mean}} = 3.87$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = 6.77$ pp; $\Delta_{\mathrm{SD,NMI}} = 2.72$ pp. | | | | | | |
| ORL / compatible | ORL_att92 | $76.00 \pm 1.60$ | $74.63 \pm 0.64$ | $88.60 \pm 1.50$ | $89.71 \pm 0.23$ | 1.37 |
| $n = 400$; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.37$ pp; $\Delta_{\mathrm{NMI}} = 1.11$ pp; $D_{\mathrm{mean}} = 1.24$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.96$ pp; $\Delta_{\mathrm{SD,NMI}} = -1.27$ pp. | | | | | | |
| UMIST / compatible | UMIST_LZUFE574_stratified_seed079 | $71.10 \pm 1.80$ | $69.97 \pm 2.27$ | $84.10 \pm 1.50$ | $85.17 \pm 1.00$ | 1.13 |
| $n = 564$; Y0, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -1.13$ pp; $\Delta_{\mathrm{NMI}} = 1.07$ pp; $D_{\mathrm{mean}} = 1.10$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.47$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.50$ pp. | | | | | | |
| BinAlpha / adapted | BinAlpha_first30 | $49.40 \pm 1.30$ | $50.13 \pm 1.39$ | $64.10 \pm 1.40$ | $61.94 \pm 1.07$ | 2.16 |
| $n = 780$; Y0, E8, $\gamma = 1$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = 0.73$ pp; $\Delta_{\mathrm{NMI}} = -2.16$ pp; $D_{\mathrm{mean}} = 1.44$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.09$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.33$ pp. | | | | | | |
| BinAlpha / compatible | BinAlpha_random_seed057 | $49.40 \pm 1.30$ | $49.54 \pm 2.71$ | $64.10 \pm 1.40$ | $63.46 \pm 1.04$ | 0.64 |
| $n = 780$; Y0, E8, $\gamma = 1e+06$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = 0.14$ pp; $\Delta_{\mathrm{NMI}} = -0.64$ pp; $D_{\mathrm{mean}} = 0.39$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 1.41$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.36$ pp. | | | | | | |
| MSRA50 / adapted | MSRA25_EBSA_16x16 | $63.20 \pm 1.10$ | $49.25 \pm 4.38$ | $72.20 \pm 1.40$ | $61.50 \pm 3.89$ | 13.95 |
| $n = 1799$; Y0, E13, $\gamma = 1e+06$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -13.95$ pp; $\Delta_{\mathrm{NMI}} = -10.70$ pp; $D_{\mathrm{mean}} = 12.32$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 3.28$ pp; $\Delta_{\mathrm{SD,NMI}} = 2.49$ pp. | | | | | | |
| YaleB / compatible | YaleB_cropped2414_nearest_Y3 | $49.80 \pm 1.60$ | $44.24 \pm 0.82$ | $46.80 \pm 1.30$ | $52.61 \pm 0.36$ | 5.81 |
| $n = 2414$; Y3, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -5.56$ pp; $\Delta_{\mathrm{NMI}} = 5.81$ pp; $D_{\mathrm{mean}} = 5.68$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.78$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.94$ pp. | | | | | | |
| USPS / compatible | USPS | $75.50 \pm 1.90$ | $66.26 \pm 1.70$ | $79.80 \pm 1.60$ | $80.86 \pm 1.20$ | 9.24 |
| $n = 9298$; Y0, E13, $\gamma = 1$, G; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -9.24$ pp; $\Delta_{\mathrm{NMI}} = 1.06$ pp; $D_{\mathrm{mean}} = 5.15$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.20$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.40$ pp. | | | | | | |

## Single native-best assignment among 50 fixed-G roundings

| Data / pool | Selected candidate | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| AR / compatible | AR_common_mask3306 | $51.30 \pm 1.50$ | 51.19 | $73.20 \pm 1.40$ | 73.34 | 0.14 |
| $n = 840$; Y0, E8, $\gamma = 1$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -0.11$ pp; $\Delta_{\mathrm{NMI}} = 0.14$ pp; $D_{\mathrm{mean}} = 0.12$; UNVERIFIED. | | | | | | |
| JAFFE / compatible | JAFFE | $81.20 \pm 1.60$ | 95.31 | $84.30 \pm 1.60$ | 94.09 | 14.11 |
| $n = 213$; Y0, E8, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = 14.11$ pp; $\Delta_{\mathrm{NMI}} = 9.79$ pp; $D_{\mathrm{mean}} = 11.95$; documented source; no pixel identity proof. | | | | | | |
| ORL / compatible | ORL_att92 | $76.00 \pm 1.60$ | 75.00 | $88.60 \pm 1.50$ | 88.60 | 1.00 |
| $n = 400$; Y0, E8, $\gamma = 0.001$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.00$ pp; $\Delta_{\mathrm{NMI}} = 0.00$ pp; $D_{\mathrm{mean}} = 0.50$; documented source; no pixel identity proof. | | | | | | |
| UMIST / compatible | UMIST_LZUFE574_stratified_seed022 | $71.10 \pm 1.80$ | 70.92 | $84.10 \pm 1.50$ | 84.36 | 0.26 |
| $n = 564$; Y0, E8, $\gamma = 0.001$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -0.18$ pp; $\Delta_{\mathrm{NMI}} = 0.26$ pp; $D_{\mathrm{mean}} = 0.22$; UNVERIFIED. | | | | | | |
| BinAlpha / adapted | BinAlpha_first30 | $49.40 \pm 1.30$ | 48.33 | $64.10 \pm 1.40$ | 62.97 | 1.13 |
| $n = 780$; Y0, E8, $\gamma = 1000$, G; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.07$ pp; $\Delta_{\mathrm{NMI}} = -1.13$ pp; $D_{\mathrm{mean}} = 1.10$; UNVERIFIED. | | | | | | |
| BinAlpha / compatible | BinAlpha_random_seed055 | $49.40 \pm 1.30$ | 49.36 | $64.10 \pm 1.40$ | 64.11 | 0.04 |
| $n = 780$; Y0, E13, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -0.04$ pp; $\Delta_{\mathrm{NMI}} = 0.01$ pp; $D_{\mathrm{mean}} = 0.03$; UNVERIFIED. | | | | | | |
| MSRA50 / adapted | MSRA25_EBSA_16x16 | $63.20 \pm 1.10$ | 56.25 | $72.20 \pm 1.40$ | 64.57 | 7.63 |
| $n = 1799$; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -6.95$ pp; $\Delta_{\mathrm{NMI}} = -7.63$ pp; $D_{\mathrm{mean}} = 7.29$; UNVERIFIED. | | | | | | |
| YaleB / compatible | YaleB_cropped2414_nearest_Y3 | $49.80 \pm 1.60$ | 44.74 | $46.80 \pm 1.30$ | 52.28 | 5.48 |

| Data / pool | Selected candidate | Paper ACC | Our ACC | Paper NMI | Our NMI | $D_{\max}$ |
|---|---|---|---|---|---|---|
| $n = 2414$; Y3, E13, $\gamma = 1000$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -5.06$ pp; $\Delta_{\mathrm{NMI}} = 5.48$ pp; $D_{\mathrm{mean}} = 5.27$; documented source; no pixel identity proof. | | | | | | |
| USPS / compatible | USPS | $75.50 \pm 1.90$ | 65.63 | $79.80 \pm 1.60$ | 81.17 | 9.87 |
| $n = 9298$; Y0, E8, $\gamma = 1$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -9.87$ pp; $\Delta_{\mathrm{NMI}} = 1.37$ pp; $D_{\mathrm{mean}} = 5.62$; documented source; no pixel identity proof. | | | | | | |

### B.5 Separate certified-search availability

The closest subset matches above belong to the locked finite historical-style IRLS track. A complete 50-start execution does not establish optimality. No comparable certified subset-search track was executed, so no certified closest winner is selected. Separate numerical references, including a continuation on the current USPS input, cannot transfer a certificate to a different saved G. Counts below describe the historical-style coverage only; native rows summarize the same 50-start configuration pool but select one assignment, with no across-run SD.

Table 7: Explicit availability by intended eligible pool and endpoint. Every row has machine status UNAVAILABLE_NO_COMPARABLE_CERTIFIED_SEARCH_COVERAGE in certified_search_status.csv; no winner is fabricated. Aliases are shown in the complete count and removed in the canonical count.

| Data | Pool | Endpoint | Complete / canonical configs | Unique inputs | Certified closest |
|---|---|---|---|---|---|
| AR | Compatible | Mean | 41640 / 41520 | 1384 | Unavailable |
| AR | Compatible | Native | 41640 / 41520 | 1384 | Unavailable |
| BinAlpha | Adapted | Mean | 60 / 60 | 2 | Unavailable |
| BinAlpha | Adapted | Native | 60 / 60 | 2 | Unavailable |
| BinAlpha | Compatible | Mean | 3000 / 3000 | 100 | Unavailable |
| BinAlpha | Compatible | Native | 3000 / 3000 | 100 | Unavailable |
| JAFFE | Compatible | Mean | 30 / 30 | 1 | Unavailable |
| JAFFE | Compatible | Native | 30 / 30 | 1 | Unavailable |
| MSRA50 | Adapted | Mean | 30 / 30 | 1 | Unavailable |
| MSRA50 | Adapted | Native | 30 / 30 | 1 | Unavailable |
| ORL | Compatible | Mean | 30 / 30 | 1 | Unavailable |
| ORL | Compatible | Native | 30 / 30 | 1 | Unavailable |
| UMIST | Compatible | Mean | 31980 / 28980 | 966 | Unavailable |
| UMIST | Compatible | Native | 31980 / 28980 | 966 | Unavailable |
| USPS | Compatible | Mean | 18 / 18 | 1 | Unavailable |
| USPS | Compatible | Native | 18 / 18 | 1 | Unavailable |
| YaleB | Compatible | Mean | 480 / 480 | 16 | Unavailable |
| YaleB | Compatible | Native | 480 / 480 | 16 | Unavailable |

### B.6 Additional rounding on locked winners

| Data | Endpoint | Select ACC | Select NMI | Confirm ACC | Confirm NMI | Status |
|---|---|---|---|---|---|---|
| AR | mean | 51.21 | 72.94 | $51.14 \pm 1.91$ | $72.88 \pm 1.03$ | COMPLETED |
| BinAlpha | mean | 50.13 | 61.94 | $50.14 \pm 1.37$ | $62.03 \pm 1.02$ | COMPLETED |
| BinAlpha | mean | 49.54 | 63.46 | $49.26 \pm 2.81$ | $63.36 \pm 1.14$ | COMPLETED |
| JAFFE | mean | 74.75 | 85.60 | $73.96 \pm 11.26$ | $84.88 \pm 5.91$ | COMPLETED |
| MSRA50 | mean | 49.25 | 61.50 | $49.67 \pm 4.06$ | $61.38 \pm 3.82$ | COMPLETED |
| ORL | mean | 74.63 | 89.71 | $74.51 \pm 0.69$ | $89.64 \pm 0.26$ | COMPLETED |
| UMIST | mean | 69.97 | 85.17 | $70.60 \pm 2.13$ | $85.51 \pm 0.85$ | COMPLETED |
| USPS | mean | 66.26 | 80.86 | $65.69 \pm 2.77$ | $79.73 \pm 2.37$ | COMPLETED |
| YaleB | mean | 44.24 | 52.61 | $44.37 \pm 0.83$ | $52.72 \pm 0.38$ | COMPLETED |
| AR | native | 51.19 | 73.34 | 51.67 | 77.42 | COMPLETED |
| BinAlpha | native | 48.33 | 62.97 | 49.62 | 63.10 | COMPLETED |
| BinAlpha | native | 49.36 | 64.11 | 50.77 | 64.58 | COMPLETED |
| JAFFE | native | 95.31 | 94.09 | 95.31 | 94.09 | COMPLETED |
| MSRA50 | native | 56.25 | 64.57 | 46.08 | 62.82 | COMPLETED |
| ORL | native | 75.00 | 88.60 | 76.50 | 89.86 | COMPLETED |

| Data | Endpoint | Select ACC | Select NMI | Confirm ACC | Confirm NMI | Status |
|---|---|---|---|---|---|---|
| UMIST | native | 70.92 | 84.36 | 60.82 | 79.17 | COMPLETED |
| USPS | native | 65.63 | 81.17 | 65.63 | 81.17 | COMPLETED |
| YaleB | native | 44.74 | 52.28 | 43.37 | 51.63 | COMPLETED |

Additional seeds check rounding on the same samples and saved G. Winners are not reselected. These intervals/SD do not establish historical identity, unseen-data generalization or label-free model selection.

### B.7 Baseline comparisons

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| JAFFE | Ridge LDA (adapted) | limit 300 | 83.85 ± 2.50 | 87.77 ± 1.53 | COMPLETED |
| Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| JAFFE | Ridge LDA–SC (adapted) | adapted_sc | 59.38 ± 9.77 | 73.72 ± 5.60 | COMPLETED |
| JAFFE | $PCA_{p_0}$–SC | pca_p0_sc | 69.00 ± 11.08 | 83.02 ± 5.86 | COMPLETED |
| JAFFE | $PCA_r$–SC | pca_r_sc | 58.64 ± 7.81 | 73.20 ± 5.04 | COMPLETED |
| JAFFE | Raw SC | raw_sc | 68.53 ± 11.15 | 83.02 ± 5.72 | COMPLETED |
| MSRA25_EBSA_16x16 | Ridge LDA (adapted) | limit 300 | 57.25 ± 4.00 | 72.24 ± 2.52 | COMPLETED |
| Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| MSRA25_EBSA_16x16 | Ridge LDA–SC (adapted) | adapted_sc | 43.61 ± 4.77 | 54.73 ± 3.91 | COMPLETED |
| MSRA25_EBSA_16x16 | $PCA_{p_0}$–SC | pca_p0_sc | 49.54 ± 5.74 | 62.89 ± 4.11 | COMPLETED |
| MSRA25_EBSA_16x16 | $PCA_r$–SC | pca_r_sc | 37.81 ± 3.64 | 49.97 ± 2.60 | COMPLETED |
| MSRA25_EBSA_16x16 | Raw SC | raw_sc | 49.14 ± 6.24 | 62.84 ± 4.72 | COMPLETED |
| ORL_att92 | Ridge LDA (adapted) | limit 300 | 75.37 ± 3.38 | 88.94 ± 1.53 | COMPLETED |
| Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| ORL_att92 | Ridge LDA–SC (adapted) | adapted_sc | 49.19 ± 3.76 | 75.66 ± 2.65 | COMPLETED |
| ORL_att92 | $PCA_{p_0}$–SC | pca_p0_sc | 55.44 ± 4.28 | 80.12 ± 2.25 | COMPLETED |
| ORL_att92 | $PCA_r$–SC | pca_r_sc | 55.29 ± 4.36 | 80.46 ± 2.21 | COMPLETED |
| ORL_att92 | Raw SC | raw_sc | 56.18 ± 4.19 | 80.47 ± 2.29 | COMPLETED |
| USPS | Ridge LDA (adapted) | limit 300 | 70.17 ± 0.11 | 68.70 ± 0.09 | COMPLETED |
| Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| USPS | Ridge LDA–SC (adapted) | adapted_sc | 57.79 ± 6.19 | 61.10 ± 2.66 | COMPLETED |
| USPS | $PCA_{p_0}$–SC | pca_p0_sc | 62.78 ± 7.95 | 75.52 ± 4.48 | COMPLETED |
| USPS | $PCA_r$–SC | pca_r_sc | 55.93 ± 4.32 | 65.98 ± 2.83 | COMPLETED |
| USPS | Raw SC | raw_sc | 61.74 ± 8.09 | 74.94 ± 4.49 | COMPLETED |
| YaleB_cropped2414 | Ridge LDA (adapted) | limit 300 | 42.12 ± 1.62 | 51.32 ± 1.22 | COMPLETED |
| Stops: MAX ITER (32), PARTITION STABLE (18); failed updates: 0; zero-success runs: 0. | | | | | |
| YaleB_cropped2414 | Ridge LDA–SC (adapted) | adapted_sc | 35.85 ± 2.62 | 50.58 ± 1.59 | COMPLETED |
| YaleB_cropped2414 | $PCA_{p_0}$–SC | pca_p0_sc | 17.74 ± 1.15 | 26.37 ± 0.64 | COMPLETED |
| YaleB_cropped2414 | $PCA_r$–SC | pca_r_sc | 15.71 ± 1.06 | 22.62 ± 0.87 | COMPLETED |
| YaleB_cropped2414 | Raw SC | raw_sc | 18.34 ± 1.28 | 26.90 ± 0.67 | COMPLETED |
| JAFFE | LDA existing | limit 100 | 64.44 ± 9.19 | 78.60 ± 5.08 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| JAFFE | LDA existing | limit 30 | 64.44 ± 9.19 | 78.60 ± 5.08 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| JAFFE | LDA existing | limit 300 | 64.44 ± 9.19 | 78.60 ± 5.08 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| MSRA25_EBSA_16x16 | LDA existing | limit 100 | 46.88 ± 5.32 | 60.43 ± 4.28 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| MSRA25_EBSA_16x16 | LDA existing | limit 30 | 46.87 ± 5.33 | 60.41 ± 4.30 | COMPLETED |
| Stops: LABELS STABLE (49), MAX ITER (1); failed updates: 0; zero-success runs: 0. | | | | | |
| MSRA25_EBSA_16x16 | LDA existing | limit 300 | 46.88 ± 5.32 | 60.43 ± 4.28 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| ORL_att92 | LDA existing | limit 100 | 56.31 ± 4.14 | 81.00 ± 1.65 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| ORL_att92 | LDA existing | limit 30 | 56.31 ± 4.14 | 81.00 ± 1.65 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| ORL_att92 | LDA existing | limit 300 | 56.31 ± 4.14 | 81.00 ± 1.65 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| USPS | LDA existing | limit 100 | 58.29 ± 6.61 | 59.93 ± 4.52 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| USPS | LDA existing | limit 30 | 58.61 ± 6.33 | 60.29 ± 4.21 | COMPLETED |
| Stops: LABELS STABLE (12), MAX ITER (38); failed updates: 0; zero-success runs: 0. | | | | | |
| USPS | LDA existing | limit 300 | 58.29 ± 6.61 | 59.93 ± 4.52 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| YaleB_cropped2414 | LDA existing | limit 100 | 9.67 ± 0.82 | 13.42 ± 1.06 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| YaleB_cropped2414 | LDA existing | limit 300 | 9.67 ± 0.82 | 13.42 ± 1.06 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| YaleB_cropped2414 | LDA existing | limit 30 | 9.67 ± 0.82 | 13.42 ± 1.06 | COMPLETED |
| Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| JAFFE | ClustPy reference | limit 300 | 82.95 ± 1.07 | 87.83 ± 0.83 | COMPLETED |
| Stops: SOURCE NMI STABLE (50); failed updates: 625; zero-success runs: 50. | | | | | |
| MSRA25_EBSA_16x16 | ClustPy reference | limit 300 | 54.19 ± 3.44 | 65.42 ± 2.47 | COMPLETED |
| Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| ORL_att92 | ClustPy reference | limit 300 | 74.22 ± 2.58 | 87.76 ± 1.11 | COMPLETED |
| Stops: MAX ITER (50); failed updates: 15000; zero-success runs: 50. | | | | | |
| USPS | ClustPy reference | limit 300 | 67.23 ± 0.14 | 65.77 ± 0.13 | COMPLETED |
| Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| YaleB_cropped2414 | ClustPy reference | limit 300 | 10.57 ± 0.75 | 14.84 ± 1.24 | COMPLETED |
| Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0. | | | | | |
| MSRA25_EBSA_16x16 | LLC | 0.0625 | 51.53 ± 0.40 | 66.90 ± 0.43 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 0.25 | 55.50 ± 1.50 | 66.52 ± 0.92 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 1.0 | 58.91 ± 2.11 | 68.28 ± 1.07 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 16.0 | 58.72 ± 2.74 | 69.12 ± 1.55 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 4.0 | 59.16 ± 2.37 | 69.04 ± 1.00 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 0.0625 | 56.01 ± 2.67 | 66.20 ± 1.22 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 0.25 | 51.80 ± 3.26 | 58.84 ± 1.56 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 1.0 | 53.73 ± 2.57 | 63.53 ± 1.83 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 16.0 | 52.15 ± 3.09 | 62.69 ± 1.98 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 4.0 | 52.52 ± 2.73 | 62.92 ± 1.96 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 0.0625 | 56.42 ± 2.20 | 64.75 ± 1.23 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 0.25 | 52.82 ± 2.55 | 62.63 ± 1.58 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| MSRA25_EBSA_16x16 | LLC | 1.0 | 52.88 ± 2.63 | 63.12 ± 1.85 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 16.0 | 51.86 ± 3.05 | 62.73 ± 1.55 | COMPLETED |
| MSRA25_EBSA_16x16 | LLC | 4.0 | 53.26 ± 2.84 | 63.28 ± 1.72 | COMPLETED |
| MSRA25_EBSA_16x16 | K-means | kmeans | 51.23 ± 4.85 | 59.81 ± 3.41 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d10 | 49.68 ± 4.36 | 56.62 ± 2.86 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d100 | 51.22 ± 4.92 | 59.77 ± 3.50 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d110 | 51.16 ± 4.91 | 59.70 ± 3.49 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d120 | 51.17 ± 4.90 | 59.71 ± 3.47 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d130 | 51.07 ± 4.83 | 59.69 ± 3.40 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d140 | 51.04 ± 4.87 | 59.62 ± 3.46 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d150 | 51.08 ± 4.84 | 59.65 ± 3.39 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d20 | 50.87 ± 5.31 | 58.73 ± 3.75 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d30 | 51.19 ± 4.93 | 59.57 ± 3.62 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d40 | 51.43 ± 4.51 | 59.68 ± 3.37 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d50 | 51.27 ± 4.73 | 59.69 ± 3.45 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d60 | 51.12 ± 4.74 | 59.50 ± 3.44 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d70 | 51.28 ± 4.89 | 59.74 ± 3.41 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d80 | 51.21 ± 4.91 | 59.76 ± 3.53 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–K-means | pca_kmeans__d90 | 51.22 ± 4.87 | 59.70 ± 3.51 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d10 | 34.55 ± 2.85 | 45.37 ± 2.60 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d100 | 49.54 ± 5.74 | 62.89 ± 4.11 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d110 | 48.93 ± 5.94 | 63.99 ± 5.10 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d120 | 48.42 ± 4.57 | 62.33 ± 3.33 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d130 | 45.67 ± 4.62 | 60.11 ± 3.56 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d140 | 46.56 ± 5.48 | 59.72 ± 4.17 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d150 | 45.73 ± 4.63 | 58.83 ± 3.40 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d20 | 43.63 ± 4.56 | 57.19 ± 3.12 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d30 | 46.00 ± 4.68 | 59.01 ± 3.38 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d40 | 45.56 ± 3.88 | 59.90 ± 2.84 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d50 | 48.51 ± 5.07 | 62.75 ± 3.42 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d60 | 46.86 ± 4.02 | 61.25 ± 2.84 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d70 | 47.86 ± 5.57 | 60.95 ± 5.14 | COMPLETED |
| MSRA25_EBSA_16x16 | PCA–SC | pca_sc__d80 | 49.23 ± 4.15 | 62.78 ± 3.67 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| MSRA25_ EBSA_16x16 | PCA–SC | pca_sc__d90 | 44.87 ± 4.91 | 59.15 ± 3.93 | COMPLETED |
| MSRA25_ EBSA_16x16 | K-means | kmeans | 51.23 ± 4.85 | 59.81 ± 3.41 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.2__ d110 | 72.21 ± 8.46 | 84.78 ± 4.19 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.2__ d30 | 72.20 ± 8.46 | 84.78 ± 4.19 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.2__ d70 | 72.21 ± 8.46 | 84.78 ± 4.19 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.5__ d110 | 74.41 ± 9.72 | 85.63 ± 4.45 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.5__ d30 | 74.45 ± 9.90 | 85.52 ± 4.58 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.5__ d70 | 74.41 ± 9.72 | 85.63 ± 4.45 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.8__ d110 | 73.06 ± 7.18 | 84.07 ± 4.05 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.8__ d30 | 73.13 ± 7.24 | 84.18 ± 4.02 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.8__ d70 | 72.85 ± 6.89 | 83.84 ± 3.96 | COMPLETED |
| JAFFE | SLPS | k5__alpha0.95_ _d70__ SKLEARN_ GREEDY_ KMEANSPP | 81.53 ± 6.24 | 86.97 ± 3.32 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.2__ d110 | 73.33 ± 8.76 | 85.46 ± 4.15 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.2__ d30 | 73.11 ± 8.97 | 85.36 ± 4.24 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.2__ d70 | 73.33 ± 8.76 | 85.46 ± 4.15 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.5__ d110 | 74.21 ± 8.75 | 85.71 ± 4.05 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.5__ d30 | 74.01 ± 8.37 | 85.59 ± 3.83 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.5__ d70 | 74.21 ± 8.75 | 85.71 ± 4.05 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.8__ d110 | 73.92 ± 8.02 | 84.72 ± 4.15 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.8__ d30 | 74.58 ± 8.28 | 85.14 ± 4.32 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.8__ d70 | 73.92 ± 8.02 | 84.72 ± 4.15 | COMPLETED |
| JAFFE | SLPS | k7__alpha0.95_ _d70__ SKLEARN_ GREEDY_ KMEANSPP | 80.85 ± 6.58 | 86.91 ± 3.58 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.2__ d110 | 52.30 ± 5.18 | 67.22 ± 3.91 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.2__ d30 | 52.62 ± 5.08 | 67.45 ± 3.85 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.2__ d70 | 52.30 ± 5.17 | 67.19 ± 3.88 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.5__ d110 | 52.10 ± 5.41 | 64.55 ± 4.36 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.5__ d30 | 52.17 ± 5.23 | 64.46 ± 4.37 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.5__ d70 | 52.16 ± 5.31 | 64.54 ± 4.27 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.8__ d110 | 51.42 ± 5.00 | 61.63 ± 3.81 | COMPLETED |
| MSRA25_ EBSA_16x16 | SLPS | k5__alpha0.8__ d30 | 51.18 ± 5.05 | 61.33 ± 3.83 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| MSRA25_EBSA_16x16 | SLPS | k5__alpha0.8__d70 | 51.43 ± 4.96 | 61.70 ± 3.77 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.2__d110 | 52.39 ± 5.28 | 67.77 ± 3.87 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.2__d30 | 52.48 ± 4.91 | 67.72 ± 3.72 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.2__d70 | 52.30 ± 5.15 | 67.69 ± 3.76 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.5__d110 | 52.10 ± 5.27 | 65.53 ± 4.35 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.5__d30 | 52.37 ± 5.40 | 65.75 ± 4.49 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.5__d70 | 52.08 ± 5.32 | 65.60 ± 4.36 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.8__d110 | 51.66 ± 5.06 | 62.58 ± 3.67 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.8__d30 | 51.71 ± 5.30 | 62.75 ± 3.88 | COMPLETED |
| MSRA25_EBSA_16x16 | SLPS | k7__alpha0.8__d70 | 51.87 ± 5.05 | 62.79 ± 3.79 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.2__d110 | 66.22 ± 3.56 | 87.15 ± 0.92 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.2__d30 | 66.65 ± 3.44 | 87.26 ± 0.95 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.2__d70 | 66.29 ± 3.62 | 87.21 ± 0.92 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.5__d110 | 65.90 ± 3.91 | 86.84 ± 1.21 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.5__d30 | 66.13 ± 3.77 | 86.91 ± 1.17 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.5__d70 | 66.07 ± 3.88 | 86.92 ± 1.19 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP | 74.68 ± 3.12 | 88.45 ± 1.19 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.8__d110 | 64.91 ± 3.18 | 85.63 ± 1.06 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.8__d30 | 66.01 ± 3.32 | 86.12 ± 1.05 | COMPLETED |
| ORL_att92 | SLPS | k5__alpha0.8__d70 | 65.20 ± 3.15 | 85.78 ± 1.10 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.2__d110 | 64.53 ± 3.72 | 85.88 ± 1.09 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.2__d30 | 64.44 ± 3.77 | 85.78 ± 1.08 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.2__d70 | 64.45 ± 3.77 | 85.90 ± 1.10 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.5__d110 | 65.19 ± 3.72 | 85.82 ± 1.10 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.5__d30 | 65.51 ± 3.77 | 85.81 ± 1.11 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.5__d70 | 65.26 ± 3.76 | 85.82 ± 1.08 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP | 74.19 ± 1.98 | 87.90 ± 0.87 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.8__d110 | 64.61 ± 3.01 | 85.38 ± 1.10 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.8__d30 | 65.57 ± 3.40 | 85.58 ± 1.14 | COMPLETED |
| ORL_att92 | SLPS | k7__alpha0.8__d70 | 64.59 ± 2.78 | 85.40 ± 1.06 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| YaleB_cropped2414 | SLPS | k5__alpha0.2__d110 | 11.83 ± 1.07 | 18.36 ± 1.01 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.2__d30 | 11.76 ± 1.02 | 18.28 ± 0.97 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.2__d70 | 11.83 ± 1.06 | 18.37 ± 0.96 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP | 12.11 ± 0.66 | 18.46 ± 0.62 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.5__d110 | 10.98 ± 0.92 | 16.51 ± 0.84 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.5__d30 | 10.83 ± 0.77 | 16.29 ± 0.75 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.5__d70 | 10.96 ± 0.87 | 16.45 ± 0.84 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.8__d110 | 9.91 ± 0.75 | 14.04 ± 0.85 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.8__d30 | 9.74 ± 0.65 | 13.90 ± 0.82 | COMPLETED |
| YaleB_cropped2414 | SLPS | k5__alpha0.8__d70 | 9.88 ± 0.70 | 14.01 ± 0.84 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.2__d110 | 11.13 ± 1.01 | 16.95 ± 0.94 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.2__d30 | 11.11 ± 1.01 | 16.92 ± 0.93 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.2__d70 | 11.13 ± 1.03 | 16.94 ± 0.93 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP | 11.24 ± 0.54 | 17.07 ± 0.44 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.5__d110 | 10.79 ± 0.89 | 16.02 ± 0.89 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.5__d30 | 10.70 ± 0.89 | 15.96 ± 0.87 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.5__d70 | 10.82 ± 0.86 | 16.05 ± 0.86 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.8__d110 | 10.09 ± 0.78 | 14.22 ± 0.84 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.8__d30 | 9.94 ± 0.73 | 14.08 ± 0.84 | COMPLETED |
| YaleB_cropped2414 | SLPS | k7__alpha0.8__d70 | 10.10 ± 0.77 | 14.21 ± 0.81 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.2__d110 | 13.07 ± 1.17 | 20.48 ± 1.11 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.2__d30 | 12.90 ± 1.10 | 20.32 ± 1.06 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.2__d70 | 13.05 ± 1.19 | 20.54 ± 1.13 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP | 14.42 ± 0.73 | 21.36 ± 0.72 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.5__d110 | 11.81 ± 0.95 | 18.17 ± 0.91 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.5__d30 | 11.51 ± 0.93 | 17.76 ± 0.82 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.5__d70 | 11.76 ± 0.93 | 18.07 ± 0.88 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.8__d110 | 10.51 ± 0.80 | 15.06 ± 0.79 | COMPLETED |
| YaleB_dengcai32 | SLPS | k5__alpha0.8__d30 | 10.18 ± 0.80 | 14.70 ± 0.88 | COMPLETED |

| Data | Method | Setting | ACC | NMI | Status |
|---|---|---|---|---|---|
| YaleB_dengcai32 | SLPS | k5__alpha0.8__d70 | $10.40 \pm 0.82$ | $14.90 \pm 0.73$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.2__d110 | $12.36 \pm 1.23$ | $18.90 \pm 1.11$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.2__d30 | $12.24 \pm 1.20$ | $18.71 \pm 1.11$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.2__d70 | $12.33 \pm 1.23$ | $18.83 \pm 1.08$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP | $13.52 \pm 0.78$ | $19.44 \pm 0.69$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.5__d110 | $11.85 \pm 1.02$ | $17.74 \pm 0.90$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.5__d30 | $11.61 \pm 0.99$ | $17.51 \pm 0.89$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.5__d70 | $11.87 \pm 0.99$ | $17.79 \pm 0.89$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.8__d110 | $10.65 \pm 0.79$ | $15.35 \pm 0.75$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.8__d30 | $10.48 \pm 0.79$ | $15.15 \pm 0.86$ | COMPLETED |
| YaleB_dengcai32 | SLPS | k7__alpha0.8__d70 | $10.66 \pm 0.83$ | $15.36 \pm 0.74$ | COMPLETED |

### B.8 USPS numerical audit on one newly frozen F/W pair

These rows use a newly generated, recoverable current USPS representation and its exact shrinkage graph. They do not identify or replace the missing historical selected F/G arrays. Every non-SC point targets unsquared E8 with undirected coefficient $2\gamma w$. All original ten stages remain visible; the two additional restarts are separate numerical attempts, not replacements for the historical finite-iteration winner or subset-search output.

The numerical audit and subset USPS representation record exactly matching current F/W content addresses. Recorded input-reference status: MATCHING_RECORDED_CURRENT_F_W_OBJECTS. This is only a comparison of current input references. Each certificate belongs to its own saved continuation G; it does not certify the locked IRLS G, establish a certified closest search result, or recover missing historical F/G arrays.

Tables 10 and 11 refer to the same stage/gamma rows. C/U denote the recorded certified/uncertified state, with a required relative gap at most $10^{-5}$ and capacity violation at most $10^{-8}$ for certification. A solver stopping status alone is not a certificate. Uncertified points cannot establish that the mathematical objective lacks clustering capacity. Missing exports or saved points stay unavailable.

Table 10: Same-target numerical evidence. Original rows show fit CPU only where recorded; extra-restart CPU includes the full numerical attempt. A dash means unavailable metadata, not zero cost.

| Stage | $\gamma$ | Primal | Dual | Rel. gap | Cert. | CPU (s) | Solver status |
|---|---|---|---|---|---|---|---|
| SC on F | – | – | – | – | – | – | NOT_APPLICABLE |
| IRLS300 | 1 | 13.577 | – | – | U | 1800.0 | CPU_LIMIT |
| Uniform | 1 | 13.644 | 13.562 | 0.006 | U | 160.0 | MAX_ITER_REACHED |
| Diagonal | 1 | 13.587 | 13.576 | 0.000864 | U | 160.5 | MAX_ITER_REACHED |
| Flow | 1 | 267.96 | 1.6424e-16 | 1 | U | 12.7 | INCOMPLETE_LINEAR_SOLVE |
| Warm restart | 1 | 13.577 | 13.577 | 2.58e-05 | U | 155.2 | MAX_ITER_REACHED |
| IRLS300 | 1000 | 388.48 | – | – | U | 11.7 | FAILED |

| Stage | $\gamma$ | Primal | Dual | Rel. gap | Cert. | CPU (s) | Solver status |
|---|---|---|---|---|---|---|---|
| Uniform | 1000 | 267.96 | 87.328 | 0.674 | U | 154.2 | MAX_ITER_ REACHED |
| Diagonal | 1000 | 267.96 | 86.357 | 0.678 | U | 142.7 | MAX_ITER_ REACHED |
| Flow | 1000 | 267.96 | 1.6424e-13 | 1 | U | 11.3 | INCOMPLETE_ LINEAR_SOLVE |
| Warm restart | 1000 | 267.96 | 120.26 | 0.551 | U | 136.5 | MAX_ITER_ REACHED |
| Extra restart | 1 | 13.577 | 13.577 | 1e-05 | C | 269.4 | PRIMAL_DUAL_ GAP_CERTIFIED |
| Extra restart | 1000 | 267.96 | 176.38 | 0.342 | U | 811.4 | MAX_ITER_ REACHED |

Clustering metrics below are percent, mean $\pm$ sample SD over the same fifty greedy K-means++/Lloyd seeds 0–49 on each saved G. Both metrics come from one complete endpoint. This is conditional rounding variability, not independent datasets or full-pipeline uncertainty. A native-best assignment is not substituted for a mean, and its single-run value receives no across-run SD.

Table 11: Paired rounding protocol on the fixed current F/W. The IRLS300 flag refers only to completion of the declared historical-style 300-step protocol, including its iterate stopping rule; it does not certify optimality. Other stages are separate algorithms/restarts.

| Stage | $\gamma$ | ACC | NMI | n | IRLS300 complete | Evaluation status |
|---|---|---|---|---|---|---|
| SC on F | – | $65.60 \pm 1.87$ | $80.31 \pm 2.13$ | 50 | – | COMPLETED |
| IRLS300 | 1 | $65.56 \pm 2.52$ | $80.23 \pm 2.34$ | 50 | no | COMPLETED |
| Uniform | 1 | $65.40 \pm 2.48$ | $80.11 \pm 2.28$ | 50 | – | COMPLETED |
| Diagonal | 1 | $65.57 \pm 2.36$ | $80.08 \pm 2.28$ | 50 | – | COMPLETED |
| Flow | 1 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |
| Warm restart | 1 | $65.56 \pm 2.52$ | $80.23 \pm 2.34$ | 50 | – | COMPLETED |
| IRLS300 | 1000 | $77.37 \pm 4.49$ | $84.84 \pm 1.56$ | 50 | no | COMPLETED |
| Uniform | 1000 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |
| Diagonal | 1000 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |
| Flow | 1000 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |
| Warm restart | 1000 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |
| Extra restart | 1 | $65.56 \pm 2.52$ | $80.23 \pm 2.34$ | 50 | – | COMPLETED |
| Extra restart | 1000 | $16.70 \pm 0.00$ | $0.00 \pm 0.00$ | 50 | – | COMPLETED |

### B.9 Fixed-representation graph evidence: G1–G3

This section uses the preregistered full ORL and cropped YaleB inputs, independently of retrospective subset winners. No best gamma, budget, graph or seed is selected here. Means and sample SD use the same fifty paired greedy K-means++/Lloyd seeds 0–49. A dash denotes unavailable evidence. C/U records certified/uncertified solver status; certification does not imply accurate clustering. The source files and hashes are listed in the generated report-input manifest.

The graph design contains 362 recorded comparisons across G1, G2 and G3. Scientific outcome and numerical qualification remain distinct.

Table 12: Graph-design coverage, including unsuccessful and unevaluated configurations. Shared mathematical problem identities are aliases, not independent measurements.

| Phase | Observation | State | Tasks |
|---|---|---|---|
| G1 | Scientific output | CASEBOOK_COMPLETE | 2 |
| G2 | Scientific output | POINT_COMPLETE | 48 |
| G3 | Scientific output | POINT_COMPLETE | 312 |

**G1: fixed gamma 1 casebooks.** Casebook correctness uses an optimal assignment separately for each partition. Error-rate means below average the fifty paired rounding outcomes; deltas are SSC minus SC. They describe the fixed data and representation, not independent test datasets.

Table 13: Matched G1 means and sample SD at fixed gamma 1; these are not selected gamma-path maxima.

| Data | SC ACC | SSC ACC | SC NMI | SSC NMI | SSC solver |
|---|---|---|---|---|---|
| ORL | $78.33 \pm 1.31$ | $78.48 \pm 1.41$ | $89.87 \pm 0.67$ | $89.89 \pm 0.80$ | PRIMAL_DUAL_GAP_ CERTIFIED |
| YaleB cropped | $21.78 \pm 0.45$ | $21.51 \pm 0.48$ | $27.73 \pm 0.58$ | $27.45 \pm 0.58$ | PRIMAL_DUAL_GAP_ CERTIFIED |

Table 14: False merges and splits use contingency-count definitions. The averaging unit is a paired rounding seed, not a sample pair treated as an independent replicate.

| Data | Pair error | SC (%) | SSC (%) | Delta (pp) | Pairs of rounds |
|---|---|---|---|---|---|
| ORL | False merge | 0.91 | 0.92 | 0.01 | 50 |
| ORL | False split | 24.37 | 24.21 | -0.16 | 50 |
| YaleB cropped | False merge | 7.78 | 7.95 | 0.17 | 50 |
| YaleB cropped | False split | 84.31 | 84.34 | 0.02 | 50 |

Table 15: Upstream A and shrinkage W0 contamination remain distinct. An unavailable A field is not replaced with W0 evidence. Fragmentation is class-induced and includes isolates.

| Data | Graph / length space | Edges | Cross edges (%) | Cross mass (%) | Excess fragments |
|---|---|---|---|---|---|
| ORL | A: input-feature geometry | 1273 | 22.31 | 11.99 | 8 |
| ORL | W0: spectral geometry | 1217 | 17.01 | 4.65 | 16 |
| YaleB cropped | A: input-feature geometry | 8510 | 58.88 | 53.60 | 438 |
| YaleB cropped | W0: spectral geometry | 7957 | 63.24 | 70.23 | 1109 |

Table 16: Predefined row-norm and weighted-degree quintiles, with original cut points and tie handling. Fractions pool the recorded sample-by-round outcomes within each fixed stratum.

| Data | Variable | Quintile | n | Wrong degree | Wrong mass | SC right/SSC wrong (%) | SC wrong/SSC right (%) |
|---|---|---|---|---|---|---|---|
| ORL | row_norm | 0 | 80 | 2.60 | 0.073 | 8.70 | 10.53 |
| ORL | row_norm | 1 | 80 | 1.06 | 0.118 | 5.78 | 6.17 |
| ORL | row_norm | 2 | 80 | 0.62 | 0.123 | 3.77 | 3.17 |
| ORL | row_norm | 3 | 80 | 0.36 | 0.160 | 4.23 | 4.03 |
| ORL | row_norm | 4 | 80 | 0.53 | 0.041 | 9.12 | 8.48 |
| ORL | weighted_ degree | 0 | 80 | 2.33 | 0.021 | 10.25 | 10.57 |
| ORL | weighted_ degree | 1 | 80 | 1.52 | 0.112 | 7.58 | 9.62 |
| ORL | weighted_ degree | 2 | 80 | 0.60 | 0.116 | 5.88 | 5.58 |
| ORL | weighted_ degree | 3 | 80 | 0.30 | 0.056 | 4.15 | 2.85 |
| ORL | weighted_ degree | 4 | 80 | 0.42 | 0.211 | 3.75 | 3.75 |
| YaleB cropped | row_norm | 0 | 483 | 5.48 | 2.305 | 1.39 | 1.67 |
| YaleB cropped | row_norm | 1 | 483 | 5.19 | 2.347 | 3.14 | 2.79 |
| YaleB cropped | row_norm | 2 | 482 | 4.91 | 2.002 | 5.10 | 5.59 |
| YaleB cropped | row_norm | 3 | 483 | 3.83 | 1.170 | 7.89 | 6.15 |
| YaleB cropped | row_norm | 4 | 483 | 1.43 | 0.363 | 2.39 | 2.36 |

| Data | Variable | Quintile | n | Wrong degree | Wrong mass | SC right/SSC wrong (%) | SC wrong/SSC right (%) |
|---|---|---|---|---|---|---|---|
| YaleB cropped | weighted_ degree | 0 | 483 | 2.58 | 0.157 | 4.94 | 4.85 |
| YaleB cropped | weighted_ degree | 1 | 483 | 3.13 | 0.618 | 4.89 | 4.19 |
| YaleB cropped | weighted_ degree | 2 | 482 | 4.09 | 1.255 | 4.23 | 4.07 |
| YaleB cropped | weighted_ degree | 3 | 483 | 4.62 | 2.070 | 3.40 | 3.42 |
| YaleB cropped | weighted_ degree | 4 | 483 | 6.42 | 4.087 | 2.45 | 2.02 |

Table 17: Deterministic first original row in each fixed seed-zero correctness category. Missing categories stay unavailable; no private image is embedded.

| Data | Seed-zero category | Original row ID | Seed |
|---|---|---|---|
| ORL | sc_correct_ssc_wrong | 1 | 0 |
| ORL | sc_wrong_ssc_correct | 271 | 0 |
| ORL | both_correct | 10 | 0 |
| ORL | both_wrong | 0 | 0 |
| YaleB cropped | sc_correct_ssc_wrong | 145 | 0 |
| YaleB cropped | sc_wrong_ssc_correct | 174 | 0 |
| YaleB cropped | both_correct | 0 | 0 |
| YaleB cropped | both_wrong | 2 | 0 |

**G2: all 48 fixed cells.** F0 is the saved raw spectral anchor; U is its row-normalized version. W0 and Wa are constructed in those respective geometries. Every graph is mass-matched to W0; equal mass does not equalize node forces or movement. Gamma zero is the anchor-only SC endpoint. Cached gamma-zero graph-arm rows are declared aliases, not independent fits.

Table 18: Complete fixed G2 path, including unavailable/failed cells. Movement is sum of row displacements divided by sum of anchor row norms; it is not Frobenius movement.

| Data | Anchor | Graph | $\gamma$ | ACC | NMI | Cert. | L21 move | Scientific outcome |
|---|---|---|---|---|---|---|---|---|
| ORL | F0 | W0 | 0 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C | 0 | POINT_COMPLETE |
| ORL | F0 | W0 | 0.001 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C | 0 | POINT_COMPLETE |
| ORL | F0 | W0 | 1 | $78.48 \pm 1.41$ | $89.89 \pm 0.80$ | C | 0.0896 | POINT_COMPLETE |
| ORL | F0 | W0 | 10 | $77.68 \pm 1.43$ | $90.44 \pm 0.56$ | C | 0.15 | POINT_COMPLETE |
| ORL | F0 | W0 | 1000 | $74.52 \pm 0.25$ | $89.66 \pm 0.18$ | U | 0.302 | POINT_COMPLETE |
| ORL | F0 | W0 | 1e+06 | $38.25 \pm 0.00$ | $66.35 \pm 0.00$ | U | 0.678 | POINT_COMPLETE |
| ORL | F0 | Wa | 0 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C | 0 | POINT_COMPLETE |
| ORL | F0 | Wa | 0.001 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C | 0 | POINT_COMPLETE |
| ORL | F0 | Wa | 1 | $77.83 \pm 1.29$ | $89.33 \pm 0.69$ | C | 0.0819 | POINT_COMPLETE |
| ORL | F0 | Wa | 10 | $77.88 \pm 1.21$ | $89.49 \pm 0.67$ | C | 0.105 | POINT_COMPLETE |
| ORL | F0 | Wa | 1000 | $78.07 \pm 1.14$ | $89.58 \pm 0.60$ | C | 0.132 | POINT_COMPLETE |
| ORL | F0 | Wa | 1e+06 | $77.74 \pm 1.32$ | $89.59 \pm 0.60$ | U | 0.148 | POINT_COMPLETE |
| ORL | U | W0 | 0 | $80.17 \pm 1.55$ | $90.89 \pm 0.64$ | C | 0 | POINT_COMPLETE |
| ORL | U | W0 | 0.001 | $80.17 \pm 1.55$ | $90.89 \pm 0.64$ | C | 0 | POINT_COMPLETE |
| ORL | U | W0 | 1 | $80.01 \pm 1.54$ | $90.94 \pm 0.58$ | C | 0.0452 | POINT_COMPLETE |
| ORL | U | W0 | 10 | $79.92 \pm 1.49$ | $90.95 \pm 0.40$ | C | 0.0882 | POINT_COMPLETE |
| ORL | U | W0 | 1000 | $74.82 \pm 0.24$ | $89.85 \pm 0.09$ | U | 0.257 | POINT_COMPLETE |
| ORL | U | W0 | 1e+06 | $72.78 \pm 0.76$ | $89.35 \pm 0.15$ | U | 0.537 | POINT_COMPLETE |
| ORL | U | Wa | 0 | $80.17 \pm 1.55$ | $90.89 \pm 0.64$ | C | 0 | POINT_COMPLETE |
| ORL | U | Wa | 0.001 | $80.17 \pm 1.55$ | $90.89 \pm 0.64$ | C | 0 | POINT_COMPLETE |
| ORL | U | Wa | 1 | $80.34 \pm 1.45$ | $90.97 \pm 0.53$ | C | 0.0213 | POINT_COMPLETE |
| ORL | U | Wa | 10 | $80.33 \pm 1.44$ | $90.94 \pm 0.57$ | C | 0.0374 | POINT_COMPLETE |
| ORL | U | Wa | 1000 | $80.01 \pm 1.51$ | $90.89 \pm 0.58$ | C | 0.0578 | POINT_COMPLETE |
| ORL | U | Wa | 1e+06 | $80.14 \pm 1.42$ | $90.88 \pm 0.61$ | U | 0.0718 | POINT_COMPLETE |
| YaleB cropped | F0 | W0 | 0 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C | 0 | POINT_COMPLETE |

| Data | Anchor | Graph | $\gamma$ | ACC | NMI | Cert. | L21 move | Scientific outcome |
|---|---|---|---|---|---|---|---|---|
| YaleB cropped | F0 | W0 | 0.001 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | F0 | W0 | 1 | $21.51 \pm 0.48$ | $27.45 \pm 0.58$ | C | 0.0976 | POINT_COMPLETE |
| YaleB cropped | F0 | W0 | 10 | $21.60 \pm 0.41$ | $27.97 \pm 0.33$ | U | 0.335 | POINT_COMPLETE |
| YaleB cropped | F0 | W0 | 1000 | $16.88 \pm 0.17$ | $24.76 \pm 0.61$ | U | 0.714 | POINT_COMPLETE |
| YaleB cropped | F0 | W0 | 1e+06 | $9.82 \pm 0.00$ | $21.40 \pm 0.00$ | U | 0.838 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 0 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 0.001 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 1 | $22.53 \pm 0.42$ | $28.08 \pm 0.53$ | C | 0.16 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 10 | $23.37 \pm 0.45$ | $28.93 \pm 0.60$ | U | 0.274 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 1000 | $3.40 \pm 0.00$ | $8.67 \pm 0.00$ | U | 0.964 | POINT_COMPLETE |
| YaleB cropped | F0 | Wa | 1e+06 | $3.40 \pm 0.00$ | $8.67 \pm 0.00$ | U | 0.964 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 0 | $25.80 \pm 0.71$ | $32.59 \pm 0.43$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 0.001 | $25.80 \pm 0.71$ | $32.59 \pm 0.43$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 1 | $25.17 \pm 0.85$ | $31.88 \pm 0.46$ | C | 0.117 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 10 | $24.56 \pm 0.50$ | $30.91 \pm 0.28$ | C | 0.393 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 1000 | $15.08 \pm 0.00$ | $30.02 \pm 0.00$ | U | 0.82 | POINT_COMPLETE |
| YaleB cropped | U | W0 | 1e+06 | $9.82 \pm 0.00$ | $21.40 \pm 0.00$ | U | 0.897 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 0 | $25.80 \pm 0.71$ | $32.59 \pm 0.43$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 0.001 | $25.80 \pm 0.71$ | $32.59 \pm 0.43$ | C | 0 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 1 | $26.11 \pm 0.77$ | $32.68 \pm 0.53$ | C | 0.0494 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 10 | $26.49 \pm 0.92$ | $32.81 \pm 0.62$ | C | 0.165 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 1000 | $28.30 \pm 0.50$ | $34.71 \pm 0.32$ | U | 0.447 | POINT_COMPLETE |
| YaleB cropped | U | Wa | 1e+06 | $3.40 \pm 0.00$ | $8.67 \pm 0.00$ | U | 0.957 | POINT_COMPLETE |

**G3: fixed primary budget and gamma.** The following rows use only the 15% addition budget and effective E8 gamma 1000. Additions come from the already frozen label-blind pool, with at most two newly incident edges per node and median positive W0 edge weight before mass matching. Oracle additions join same-class components after cross-class deletion and are reused unchanged in add-only and both arms. Other gammas and the 5% budget remain sensitivity records in the full CSVs, not alternative promotion gates.

Table 19: Fixed primary G3 arms. Blind controls retain exactly the Oracle deletion mask and randomize only additions. Same-class random additions do not prefer separate components. Fully randomized deletion/addition is a separate supplementary control. Infeasible or unrun controls retain their status.

| Data | Arm / control | ACC | NMI | Cert. | L21 move | Excess fragments | Outcome |
|---|---|---|---|---|---|---|---|
| ORL | No edits | $74.52 \pm 0.25$ | $89.66 \pm 0.18$ | U | 0.302 | 16 | POINT_ COMPLETE |
| ORL | Oracle delete | $82.92 \pm 1.22$ | $92.95 \pm 0.37$ | C | 0.183 | 16 | POINT_ COMPLETE |
| ORL | Oracle add | $68.57 \pm 0.21$ | $90.40 \pm 0.04$ | U | 0.457 | 1 | POINT_ COMPLETE |
| ORL | Oracle delete+completion | $93.34 \pm 1.15$ | $97.98 \pm 0.33$ | C | 0.302 | 1 | POINT_ COMPLETE |
| ORL | Fixed delete + blind add 400 | $60.92 \pm 0.22$ | $86.70 \pm 0.18$ | C | 0.4 | 16 | POINT_ COMPLETE |
| ORL | Fixed delete + blind add 401 | $73.80 \pm 0.67$ | $91.57 \pm 0.28$ | C | 0.355 | 15 | POINT_ COMPLETE |
| ORL | Fixed delete + blind add 402 | $70.71 \pm 0.17$ | $89.79 \pm 0.18$ | C | 0.39 | 15 | POINT_ COMPLETE |
| ORL | Fixed delete + same-class random 400 | $87.19 \pm 1.24$ | $95.28 \pm 0.37$ | C | 0.244 | 11 | POINT_ COMPLETE |
| ORL | Fixed delete + same-class random 401 | $85.97 \pm 1.00$ | $94.44 \pm 0.24$ | C | 0.226 | 13 | POINT_ COMPLETE |
| ORL | Fixed delete + same-class random 402 | $86.24 \pm 1.34$ | $94.90 \pm 0.31$ | C | 0.228 | 13 | POINT_ COMPLETE |
| ORL | Fully random delete/add 400 | $57.62 \pm 0.62$ | $83.55 \pm 0.35$ | U | 0.553 | 15 | POINT_ COMPLETE |
| ORL | Fully random delete/add 401 | $68.11 \pm 0.44$ | $87.83 \pm 0.19$ | U | 0.516 | 19 | POINT_ COMPLETE |
| ORL | Fully random delete/add 402 | $61.64 \pm 0.96$ | $84.51 \pm 0.40$ | U | 0.568 | 17 | POINT_ COMPLETE |
| YaleB cropped | No edits | $16.88 \pm 0.17$ | $24.76 \pm 0.61$ | U | 0.714 | 1109 | POINT_ COMPLETE |
| YaleB cropped | Oracle delete | $23.17 \pm 0.64$ | $30.63 \pm 0.71$ | U | 0.173 | 1109 | POINT_ COMPLETE |
| YaleB cropped | Oracle add | $11.64 \pm 0.49$ | $17.43 \pm 0.83$ | C | 0.869 | 53 | POINT_ COMPLETE |
| YaleB cropped | Oracle delete+completion | $54.04 \pm 1.85$ | $80.29 \pm 0.84$ | U | 0.764 | 53 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + blind add 400 | $6.04 \pm 0.07$ | $11.96 \pm 0.20$ | U | 0.762 | 1000 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + blind add 401 | $6.03 \pm 0.10$ | $11.63 \pm 0.21$ | U | 0.76 | 1023 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + blind add 402 | $5.81 \pm 0.10$ | $11.58 \pm 0.19$ | U | 0.752 | 1016 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + same-class random 400 | $41.48 \pm 1.75$ | $51.96 \pm 0.93$ | C | 0.532 | 661 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + same-class random 401 | $42.25 \pm 1.71$ | $53.22 \pm 0.79$ | C | 0.538 | 640 | POINT_ COMPLETE |
| YaleB cropped | Fixed delete + same-class random 402 | $40.96 \pm 1.47$ | $51.36 \pm 0.71$ | C | 0.517 | 684 | POINT_ COMPLETE |
| YaleB cropped | Fully random delete/add 400 | $4.20 \pm 0.05$ | $9.70 \pm 0.11$ | U | 0.936 | 1376 | POINT_ COMPLETE |
| YaleB cropped | Fully random delete/add 401 | $4.20 \pm 0.05$ | $10.02 \pm 0.09$ | U | 0.936 | 1368 | POINT_ COMPLETE |
| YaleB cropped | Fully random delete/add 402 | $4.60 \pm 0.06$ | $10.57 \pm 0.13$ | U | 0.933 | 1380 | POINT_ COMPLETE |

Table 20: Preregistered promotion checks. Certification and activity are required for delete and both; blind-control certificate flags remain visible but are not an additional hidden pass condition. At least one development dataset must pass.

| Data | Locked gate condition | Threshold | Satisfied | Decision |
|---|---|---|---|---|
| ORL | Delete and both certified | gap <= 1e-5; capacities <= 1e-8 | yes | PASSED |
| ORL | Delete and both active | normalized L21 > 1e-6 | yes | PASSED |
| ORL | Both minus delete ACC | mean gain >= 1 pp | yes | PASSED |
| ORL | Both minus delete NMI | mean gain >= 0 pp | yes | PASSED |
| ORL | Class-induced excess fragments | reduction >= 20 percent | yes | PASSED |
| ORL | ACC CI versus delete | lower endpoint > 0 pp | yes | PASSED |
| ORL | ACC CI versus mean blind additions | lower endpoint > 0 pp | yes | PASSED |
| ORL | Recorded excess-fragment reduction | 93.75 percent | – | PASSED |
| YaleB cropped | Delete and both certified | gap <= 1e-5; capacities <= 1e-8 | no | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | Delete and both active | normalized L21 > 1e-6 | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | Both minus delete ACC | mean gain >= 1 pp | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | Both minus delete NMI | mean gain >= 0 pp | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | Class-induced excess fragments | reduction >= 20 percent | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | ACC CI versus delete | lower endpoint > 0 pp | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | ACC CI versus mean blind additions | lower endpoint > 0 pp | yes | NOT_EVALUABLE_UNCERTIFIED |
| YaleB cropped | Recorded excess-fragment reduction | 95.22 percent | – | NOT_EVALUABLE_UNCERTIFIED |

Table 21: Stored paired percentile intervals: 10,000 resamples, bootstrap seed 424242. The three random-control outcomes are averaged within each rounding seed before the contrast. Only ACC intervals versus deletion and the mean blind control enter the gate; supplementary controls and NMI intervals are reported without redefining it.

| Data | Fixed contrast | Metric | Mean delta (pp) | 95% lower | 95% upper |
|---|---|---|---|---|---|
| ORL | Both minus delete | ACC | 10.41 | 9.92 | 10.91 |
| ORL | Both minus delete | NMI | 5.02 | 4.87 | 5.18 |
| ORL | Both minus fixed-delete / blind-add mean | ACC | 24.86 | 24.56 | 25.17 |
| ORL | Both minus fixed-delete / blind-add mean | NMI | 8.62 | 8.53 | 8.71 |
| ORL | Both minus fixed-delete / same-class-random mean | ACC | 6.87 | 6.53 | 7.23 |
| ORL | Both minus fixed-delete / same-class-random mean | NMI | 3.10 | 3.00 | 3.21 |
| ORL | Both minus fully random edits mean | ACC | 30.88 | 30.58 | 31.19 |
| ORL | Both minus fully random edits mean | NMI | 12.68 | 12.57 | 12.79 |
| YaleB cropped | Both minus delete | ACC | 30.87 | 30.39 | 31.34 |
| YaleB cropped | Both minus delete | NMI | 49.66 | 49.34 | 49.97 |
| YaleB cropped | Both minus fixed-delete / blind-add mean | ACC | 48.07 | 47.57 | 48.57 |

| Data | Fixed contrast | Metric | Mean delta (pp) | 95% lower | 95% upper |
|---|---|---|---|---|---|
| YaleB cropped | Both minus fixed-delete / blind-add mean | NMI | 68.57 | 68.34 | 68.81 |
| YaleB cropped | Both minus fixed-delete / same-class-random mean | ACC | 12.47 | 11.92 | 13.04 |
| YaleB cropped | Both minus fixed-delete / same-class-random mean | NMI | 28.11 | 27.87 | 28.36 |
| YaleB cropped | Both minus fully random edits mean | ACC | 49.70 | 49.19 | 50.20 |
| YaleB cropped | Both minus fully random edits mean | NMI | 70.20 | 69.96 | 70.44 |

### B.10 Conditional EnSC adaptation: capability and locked operation

The primary adaptation is the fixed $\tau = .95$ EnSC graph used only as the fusion graph for anchor F0. The source coefficient parameter is 50; graphs are matched to W0 total mass. Capability rows are label-assisted oracles: one complete configuration is selected by mean ACC, then NMI, then stable ID. The primary graph has five declared SSC gamma values; original SSC receives the frozen 15-value path. Controls remain separate named families. The locked operating row uses source $50/\tau = .95$/SSC gamma 1 without choosing parameters from labels. Both kinds of row use previously examined benchmarks; neither is an unseen-data test.

Table 22: Conditional G4/G5 results, percent mean plus sample SD over 50 matched greedy one-start roundings. Missing rows stay unavailable. C/U describe certified/uncertified fusion solves; native spectral controls have no SSC gap certificate. Search completeness and comparison validity are separate from completion of an individual selected row.

| Data | Method / track | $\gamma$ | ACC | NMI | Cert. / solver |
|---|---|---|---|---|---|
| ORL | EnSC $\tau = .95$ + SSC (oracle) | 1 | $78.98 \pm 1.26$ | $91.40 \pm 0.43$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| ORL | EnSC $\tau = .95$ + SSC (locked) | 1 | $78.98 \pm 1.26$ | $91.40 \pm 0.43$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| ORL | Fair SC: U (oracle) | 0 | $80.17 \pm 1.55$ | $90.89 \pm 0.64$ | – / ANALYTIC_G_EQUALS_F |
| ORL | Original SSC (oracle) | 1 | $78.48 \pm 1.41$ | $89.89 \pm 0.80$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| ORL | Angular SSC (oracle) | 0.001 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C / ANALYTIC_G_EQUALS_F |
| ORL | Sparse $\tau = 1$ + SSC (oracle) | 1 | $79.16 \pm 1.17$ | $91.32 \pm 0.52$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| ORL | Wider-kNN SSC (oracle) | 0.001 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C / ANALYTIC_G_EQUALS_F |
| ORL | Matched random SSC (oracle) | 0.001 | $78.33 \pm 1.31$ | $89.87 \pm 0.67$ | C / ANALYTIC_G_EQUALS_F |
| ORL | Native EnSC $\tau = .95$ (oracle) | – | $81.68 \pm 1.65$ | $91.68 \pm 0.65$ | – / NATIVE_SPECTRAL_CONTROL |
| ORL | Native EnSC $\tau = 1$ (oracle) | – | $81.62 \pm 2.08$ | $91.55 \pm 0.72$ | – / NATIVE_SPECTRAL_CONTROL |
| JAFFE | EnSC $\tau = .95$ + SSC (oracle) | 1 | $96.41 \pm 0.36$ | $96.08 \pm 0.54$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| JAFFE | EnSC $\tau = .95$ + SSC (locked) | 1 | $96.41 \pm 0.36$ | $96.08 \pm 0.54$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| JAFFE | Fair SC: U (oracle) | 0 | $96.71 \pm 0.00$ | $96.39 \pm 0.00$ | – / ANALYTIC_G_EQUALS_F |
| JAFFE | Original SSC (oracle) | 10 | $95.31 \pm 0.00$ | $94.14 \pm 0.02$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| JAFFE | Angular SSC (oracle) | 1e+06 | $96.24 \pm 0.00$ | $95.52 \pm 0.00$ | U / MAX_ITER_REACHED |
| JAFFE | Sparse $\tau = 1$ + SSC (oracle) | 1 | $96.36 \pm 0.31$ | $95.99 \pm 0.46$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| JAFFE | Wider-kNN SSC (oracle) | 1000 | $95.31 \pm 0.00$ | $94.16 \pm 0.00$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| JAFFE | Matched random SSC (oracle) | 0.001 | $95.31 \pm 0.00$ | $94.13 \pm 0.02$ | C / ANALYTIC_G_EQUALS_F |

| Data | Method / track | $\gamma$ | ACC | NMI | Cert. / solver |
|---|---|---|---|---|---|
| JAFFE | Native EnSC $\tau = .95$ (oracle) | – | $99.06 \pm 0.00$ | $98.34 \pm 0.00$ | – / NATIVE_SPECTRAL_CONTROL |
| JAFFE | Native EnSC $\tau = 1$ (oracle) | – | $98.59 \pm 0.00$ | $97.78 \pm 0.00$ | – / NATIVE_SPECTRAL_CONTROL |
| YaleB | EnSC $\tau = .95$ + SSC (oracle) | 1 | $27.66 \pm 0.74$ | $36.99 \pm 0.56$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| YaleB | EnSC $\tau = .95$ + SSC (locked) | 1 | $27.66 \pm 0.74$ | $36.99 \pm 0.56$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| YaleB | Fair SC: U (oracle) | 0 | $25.80 \pm 0.71$ | $32.59 \pm 0.43$ | – / ANALYTIC_G_EQUALS_F |
| YaleB | Original SSC (oracle) | 0.001 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C / ANALYTIC_G_EQUALS_F |
| YaleB | Angular SSC (oracle) | 10 | $23.37 \pm 0.45$ | $28.93 \pm 0.60$ | U / MAX_ITER_REACHED |
| YaleB | Sparse $\tau = 1$ + SSC (oracle) | 1 | $27.59 \pm 0.80$ | $37.01 \pm 0.51$ | C / PRIMAL_DUAL_GAP_CERTIFIED |
| YaleB | Wider-kNN SSC (oracle) | 0.001 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C / ANALYTIC_G_EQUALS_F |
| YaleB | Matched random SSC (oracle) | 0.001 | $21.78 \pm 0.45$ | $27.73 \pm 0.58$ | C / ANALYTIC_G_EQUALS_F |
| YaleB | Native EnSC $\tau = .95$ (oracle) | – | $61.09 \pm 1.17$ | $66.49 \pm 0.91$ | – / NATIVE_SPECTRAL_CONTROL |
| YaleB | Native EnSC $\tau = 1$ (oracle) | – | $60.48 \pm 1.37$ | $66.04 \pm 0.95$ | – / NATIVE_SPECTRAL_CONTROL |
| USPS | EnSC $\tau = .95$ + SSC (oracle) | 0.001 | $65.60 \pm 1.87$ | $80.31 \pm 2.13$ | C / ANALYTIC_G_EQUALS_F |
| USPS | EnSC $\tau = .95$ + SSC (locked) | 1 | $52.27 \pm 3.95$ | $69.73 \pm 1.55$ | U / MAX_ITER_REACHED |
| USPS | Fair SC: U (oracle) | 0 | $68.36 \pm 3.64$ | $83.44 \pm 0.89$ | – / ANALYTIC_G_EQUALS_F |
| USPS | Original SSC (oracle) | 30 | $66.53 \pm 2.39$ | $81.33 \pm 1.20$ | U / MAX_ITER_REACHED |
| USPS | Angular SSC (oracle) | 10 | $66.97 \pm 3.23$ | $81.95 \pm 0.85$ | U / MAX_ITER_REACHED |
| USPS | Sparse $\tau = 1$ + SSC (oracle) | 0.001 | $65.60 \pm 1.87$ | $80.31 \pm 2.13$ | C / ANALYTIC_G_EQUALS_F |
| USPS | Wider-kNN SSC (oracle) | 10 | $67.11 \pm 2.51$ | $81.87 \pm 0.83$ | U / MAX_ITER_REACHED |
| USPS | Matched random SSC (oracle) | 0.001 | $65.60 \pm 1.87$ | $80.31 \pm 2.13$ | C / ANALYTIC_G_EQUALS_F |
| USPS | Native EnSC $\tau = .95$ (oracle) | – | $49.51 \pm 2.22$ | $60.82 \pm 1.19$ | – / NATIVE_SPECTRAL_CONTROL |
| USPS | Native EnSC $\tau = 1$ (oracle) | – | $49.32 \pm 2.00$ | $60.73 \pm 1.08$ | – / NATIVE_SPECTRAL_CONTROL |

The native EnSC controls cluster each learned affinity in its own spectral space, with an explicit seeded eigensolver and the same fifty greedy one-start rounds. This differs from the source constructor default of 20 K-means starts. Gains over original SC/SSC must not be credited to SSC shrinkage when the native affinity clustering already explains them. The following differences compare the selected primary $\tau = .95$ + SSC row against matched native $\tau = .95$ EnSC, when both complete rows exist. They are descriptive differences after capability selection, not unbiased intervals or matched-movement evidence.

| Data | $\Delta$ACC vs native EnSC (pp) | $\Delta$NMI vs native EnSC (pp) |
|---|---|---|
| ORL | -2.70 | -0.29 |
| JAFFE | -2.65 | -2.26 |
| YaleB | -33.43 | -29.50 |
| USPS | 16.09 | 19.49 |

**Separate capability and locked-operation targets.** Each formal row requires all four prescribed, qualified datasets with equal weights. ACC is compared with the stronger fair-SC/original-SSC baseline on each dataset; the two NMI contrasts retain those same baselines separately. Capability

uses label-assisted parameter selection. Locked operation fixes source coefficient 50, $\tau = .95$ and SSC $\gamma = 1$ without choosing those parameters from labels. Both tracks are evaluated against the same label-assisted baseline selections, on previously examined benchmarks. Neither success criterion substitutes for the other.

Capability conclusion status: INCOMPLETE_FOUR_BENCHMARK_COMPARISON. Its formal macro and practical-target decision remain NA; partial datasets or unresolved certification/control coverage are not averaged into a result.

Locked-operation conclusion status: INCOMPLETE_FOUR_BENCHMARK_COMPARISON. Its formal macro and practical-target decision remain NA; partial datasets or unresolved certification/control coverage are not averaged into a result.

Table 24: Four-benchmark targets evaluated separately, in percentage points. Each target requires average ACC gain at least 2 pp and no average NMI loss against either fixed comparator. NA remains unavailable, not a zero or a negative result.

| Track | Qualification | ΔACC strong | ΔNMI SC | ΔNMI SSC | Target met |
|---|---|---|---|---|---|
| Oracle capability | Unavailable | NA | NA | NA | NA |
| Locked $\gamma = 1$ | Unavailable | NA | NA | NA | NA |

The parameter-selection-gap flag is unavailable because both formal, qualified four-benchmark target decisions are required. No parameter-selection diagnosis is inferred from a partial track.

Table 25: Per-dataset selected-capability contrasts, in percentage points. Unqualified rows are diagnostic observations only and do not enter a formal macro conclusion. Negative values and losses remain visible.

| Data | Complete/intended | ΔACC strong | ΔNMI SC | ΔNMI SSC | Qualification |
|---|---|---|---|---|---|
| ORL | 44/44 | -1.20 | 0.51 | 1.51 | unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN |
| ORL_att92: observed ACC loss exceeds 1 percentage point against the stronger fair baseline; qualification above still applies. | | | | | |
| JAFFE | 44/44 | -0.30 | -0.31 | 1.94 | unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN |
| YaleB | 44/44 | 1.86 | 4.39 | 9.26 | unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN |
| USPS | 44/44 | -2.76 | -3.12 | -1.02 | unqualified / INCOMPLETE_OR_UNQUALIFIED_REQUIRED_DESIGN |
| USPS: observed ACC loss exceeds 1 percentage point against the stronger fair baseline; qualification above still applies. | | | | | |

### B.11 Coverage, solver qualification and cost

Complete-config input coverage means at least one completed SSC configuration on an input, not a finished search design. 2482 unique inputs yielded 74436 complete SSC input/configuration pairs; 2480 inputs completed all 30 SSC points and 2480 completed all 30 SSC plus three matched-SC points.

| Data | Intended aliases | Unique inputs | Measured inputs | Complete configurations |
|---|---|---|---|---|
| AR | 3537 | 3533 | 1385 | 41670 |
| JAFFE | 1 | 1 | 1 | 30 |
| ORL | 1 | 1 | 1 | 30 |
| UMIST | 1073 | 971 | 971 | 32190 |
| BinAlpha | 104 | 104 | 104 | 3120 |
| MSRA50 | 1 | 1 | 1 | 30 |
| YaleB | 16 | 16 | 16 | 480 |
| USPS | 3 | 3 | 3 | 66 |

| Execution status | Intended combinations |
|---|---|
| COMPLETED | 74436 |
| COMPLETED_ALIAS | 3180 |
| FAILED_COMPLETION_PRESERVED | 18 |
| FAILED_ORIGINAL_PRESERVED | 6 |
| NOT_RUN | 64440 |

The recorded scientific-campaign charge is 31.4130 CPU-core hours, including conservative accounting caps for metadata operations. Experiments used at most 2 single-thread workers and an aggregate RSS target of 8.0 GiB. Per-fit measured costs and numerical qualifications are reported separately from clustering performance.

| Experiment group | CPU seconds |
|---|---|
| G1 | 50.6 |
| G2 | 1028.5 |
| G3 | 3535.9 |
| G4_G5 | 8427.0 |
| G4_profile | 150.4 |
| aggregation | 3786.8 |
| confirmation | 11.7 |
| mandatory_baseline | 6852.2 |
| mandatory_repair | 2269.9 |
| mandatory_usps | 3840.2 |
| mandatory_usps_evaluation | 11.3 |
| profile | 838.8 |
| subset | 81380.5 |
| validation | 902.9 |

**Graph criterion.** Oracle intervention criterion: PASSED. A formal four-benchmark performance conclusion requires complete, qualified controls on all four datasets.

**Generated-input provenance.** Tables and figures are generated from the authoritative committed CSV/JSON records only. The input hash manifest is `results/submission_refresh/report_input_manifest.json`. Generated snapshot date: 2026-11-10.

## C Additional protocol and implementation detail

The source/subset catalogue fixes AR seven-of-fourteen rules, UMIST deduplication/omission hypotheses and BinAlpha 30-per-letter draws with explicit original indices. Diagnostic parents cannot compete as equivalent paper reconstructions. ORL/JAFFE membership is not arbitrarily searched. Yale source, resize interpolation and Y0–Y3 normalization factors remain separate; USPS uses all 9,298 rows in compatible experiments. Native 16-by-16 MSRA is labelled as an adapted-resolution source and is not upsampled to claim recovery of the paper's 32-by-32 data. A changed subset always rebuilds both graphs and both representations. Complete coverage, locked selections and confirmation records, rather than numerical closeness alone, determine the reported scope.

The instrumented ClustPy port pins commit `4ac9d00685efc7432800684f5bfac0f01af59172`[1]. It preserves failed-update continuation and both the actually used and source-returned projections. Scoped NumPy seeding of the source PCA is the disclosed determinism modification. The separately named PCA-preconditioned ridge LDA-Kmeans→SC adaptation uses centered float64 input, numerical rank tolerance $\epsilon \max(n,p)\sigma_{\max}$, $p_0 = \min(100, p, n-c, \mathrm{rank}(X_c))$, $r = \min(c-1, p_0)$ and ridge $\eta = 10^{-6}\mathrm{tr}(S_t)/p_0$. Covariance scatter uses divisor $n-1$; inner K-means has ten starts, the outer cap is 300, and stopping compares partitions up to label permutation. The actually clustered projection is returned. Raw SC, $\mathrm{PCA}_{p_0}$ SC and $\mathrm{PCA}_r$ SC controls use one RANDOM Lloyd start per matched external seed; direct learned-label versus downstream SC contrasts therefore also reflect their

[1] `https://github.com/collinleiber/ClustPy/blob/4ac9d00685efc7432800684f5bfac0f01af59172/clustpy/centroid/ldakmeans.py`; the original source, BSD-3-Clause notice and exact hashes are retained.

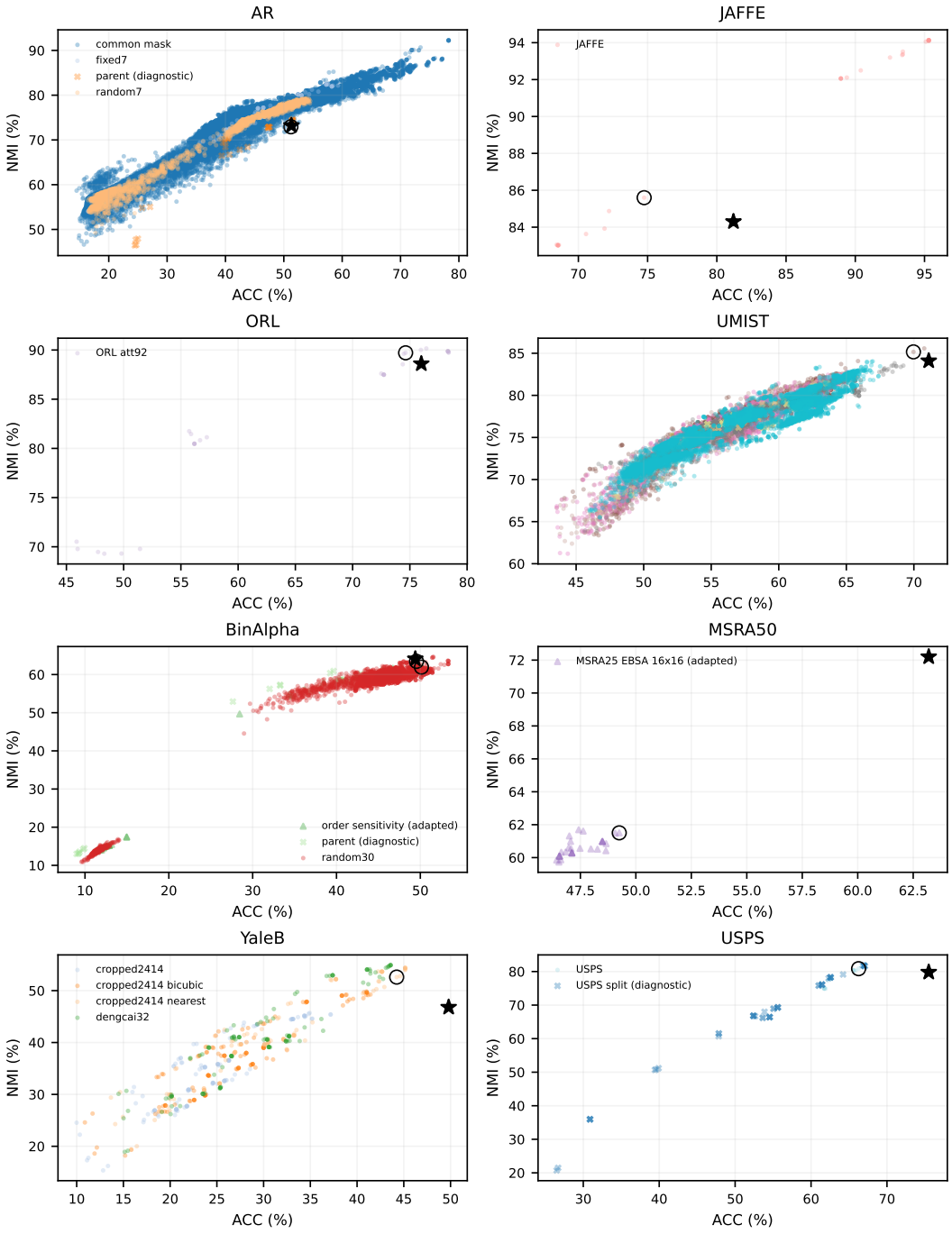



Figure 2: SSC search distribution for the mean/sample-SD endpoint. Points are complete candidate/configuration records; colors distinguish source/subset families (YaleB colors separate source/resize versions), triangles are adapted rows, crosses are diagnostic rows, stars are published SSC targets and outlines mark the already locked selections within each eligible pool. Eligibility pools and actual locked selections are reported in the tables. More candidates can improve a retrospective match without identifying historical data.

prescribed rounding budgets. Neither this adaptation nor a degenerate reference execution recovers historical LDA-SC.

The diagonal PDHG formulation uses $K = \mathrm{diag}(\sqrt{w})B$ and maps its dual back to original capacities $2\gamma w$; independent small convex references validate the target. Agreement between numerical solvers alone is not a certificate. Orthogonal eigenspace rotations preserve Euclidean distances, while a changed cutoff eigenspace or row normalization can change clustering. Edge deletion need not preserve $G = F$: on an eight-node cycle at $\gamma = 0.6$, removing one edge destroys force cancellation. The separately valid incident-mass sufficient bound is $2\gamma \max_i \sum_j w_{ij} \leq 1$.

G3 holds the full development $F_0$ fixed and uses a label-blind top-20-neighbor addition pool for oracle evaluation. The fixed 15%/gamma-1000 gate requires certified active deletion and deletion-plus-completion solves, at least one percentage point paired ACC gain, nondecreasing NMI, at least 20% fewer excess class fragments, and positive conditional bootstrap intervals against deletion and fixed oracle deletion plus mean blind random additions. Fully randomized deletion/addition and same-class random additions remain separate controls. The labelled intervention diagnoses available graph information; it is not a label-free algorithm or an unseen-data test. Subsequent EnSC capability and locked-operation targets require their own complete four-benchmark qualification; neither can be inferred from the gate alone.

The EnSC port targets the true elastic-net coefficient objective and is checked against independent small convex references. Its row-wise KKT diagnostics remain separate from SSC primal–dual certificates: a C/U fusion mark does not certify coefficient optimality. On USPS, 969 of 9,298 tau-0.95 coefficient rows fail the declared KKT diagnostic, despite source screening/objective stops. The fixed gamma-1 fusion gap is $2.8042 \times 10^{-5}$, above the certification threshold. Original-SSC gamma 30, its capability choice, is also uncertified. The USPS matched-movement comparison returns the degenerate zero-movement match $G = F$; ORL/JAFFE/YaleB have no certified movement bracket. These observations do not identify a topology-specific benefit at equal nonzero shrinkage.

## D Nonconvex fusion: scope and negative evidence

Truncated fusion is established in PRclust [Pan et al., 2013, Wu et al., 2016]; distance-dependent robust pairwise fusion is central to Shah–Koltun RCC [Shah and Koltun, 2017]. These differ from Sun et al.'s resistant convex clustering [Sun et al., 2025]. The SSC adaptation does not claim invention of these penalties.

For two concentrated groups of sizes $m < n$, separation $d$ and aggregate attraction $\lambda$, the unsquared-fidelity linear-fusion optimum is $d\min(m, \lambda)$. With capped penalty $\min(r, \tau)$ and $d > \tau$, it is $\min(md, \lambda\tau)$. Thus $m < \lambda < md/\tau$ gives fusion for the linear objective and separation for the capped objective. Equality can admit multiple minimizers. These two-representative statements are not arbitrary-graph clustering guarantees.

A separately enumerated five-point graph has linear optimum 8 and capped optimum 3. The local capped algorithm obtains 3 from F but 8 from the original SSC output: accurate inner convex solves do not certify a global nonconvex solution. Synthetic imbalanced-bridge experiments also show the limitation of damage avoidance: SSC averages 99.156 ACC/96.725 NMI while capped fusion, static screening and squared fidelity recover 100/100, which SC already achieves. Capping is therefore neither necessary in that example nor evidence of additional clustering capacity over SC. Historical synthetic and secondary geometry results remain distinct from the present graph-adaptation comparison.

The mathematical and empirical observations motivate a narrow conclusion: representation motion, robustness or objective improvement alone is insufficient. A useful extension must retain both ACC and NMI improvements over appropriate SC, SSC and affinity-only controls, with numerical qualifications attached to the exact saved outputs.